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Faculty of Economics and Administration
Department of Management Information System
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Manufacturing Community Database System

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Table of Contents

1.Executive Summary 5

2.Conceptual Database Diagram Scenario 6

2.1Entity – Relationship Diagram 7

2.2ERD Documentation 8

3.Logical Database Design 10

3.1Relational Model 10

3.2Normalization 11

4.Database Implementation 17

5.Database Implementation in MySQL 20

5.1 Tables creation 20

5.2 Insert into Tables 28

5.3 Table Modification 35

6.Querying in MySQL 36

6.1 SQL Reports 36

6.2 SQL Data Removal Statement 40

Appendix 42

List of figures

Figure 1.1 Entity Relationship Diagram (ERD)……07
Figure 2.1 Relational Model Diagram ……10
Figure 3.2.1 Normalization ……11
Figure 3.2.2 Normalization ……11
Figure 3.2.3 Normalization……12
Figure 3.2.4 Normalization……12
Figure 3.2.5 Normalization……12
Figure 3.2.6 Normalization……13
Figure 3.2.7 Normalization……13
Figure 3.2.8 Normalization ……13
Figure 3.2.9 Normalization……14
Figure 3.2.10 Normalization……14
Figure 3.2.11 Normalization……14
Figure 3.2.12 Normalization……15
Figure 3.2.13 Normalization……15
Figure 3.2.14 Normalization……15
Figure 3.2.15 Normalization……16
Figure 3.2.16 Normalization……16
Figure 3.2.17 Normalization……16
Figure 5.1-1: Tables creation……20
Figure 5.1-2: Tables creation……21
Figure 5.1-3: Tables creation……21
Figure 5.1-4: Tables creation……22
Figure 5.1-5: Tables creation……22
Figure 5.1-6: Tables creation……23
Figure 5.1-7: Tables creation……23
Figure 5.1-8: Tables creation……24
Figure 5.1-9: Tables creation……24
Figure 5.1-10: Tables creation……24
Figure 5.1-11: Tables creation……25
Figure 5.1-12: Tables creation……25
Figure 5.1-13: Tables creation……26
Figure 5.1-14: Tables creation……26
Figure 5.1-15: Tables creation……26
Figure 5.1-16: Tables creation……27
Figure 5.1-17: Tables creation……27
Figure 5.2.1 Insert into Department table……28
Figure 5.2.2 Insert into Member table……28
Figure 5.2.3 Insert into Attendee table……29
Figure 5.2.4 Insert into Event table……29
Figure 5.2.5 Insert into Participation table……30
Figure 5.2.6 Insert into Event-Attendee table……30
Figure 5.2.7 Insert into Instructor table……31

Figure 5.2.8 Insert into Event-Instructor table……31
Figure 5.2.9 Insert into Instructor _ACC table……...31
Figure 5.2.10 Insert into Sponsor table……...32
Figure 5.2.11 Insert into Event- Sponsor table……...32
Figure 5.2.12 Insert into Announcement table……...32
Figure 5.2.13 Insert into Member-Email table……...33
Figure 5.2.14 Insert into Member-Achievement table……...33
Figure 5.2.15 Insert into Department-Event table……34
Figure 5.2.16 Insert into Department-Achievement table……34
Figure 5.2.17 Insert into Feedback table……...35
Figure 5.3: Table modification……..35
Figure 6.1 Report 1……36
Figure 6.2 Report 2 ……...37
Figure 6.3 Report 3 ……...37
Figure 6.4 Report 4……38
Figure 6.5 Report 5……38
Figure 6.6 Report 6……....39
Figure 6.7 Report 7……....39
Figure 6.2.1 Remove data from table……40
Figure 6.2.2 Event-Sponsor Before remove data……..40
Figure 6.2.3 Sponsor Before remove data……40
Figure 6.2.4 Event-Sponsor After remove data……41
Figure 6.2.5 Sponsor After remove data……....41

Table of table

Table 4.1 table Instances……...17
Table 4.2 table Instances……...17
Table 4.3 table Instances ……17
Table 4.4 table Instances ……17
Table 4.5 table Instances ……18
Table 4.6 table Instances ……18
Table 4.7 table Instances ……18
Table 4.8 table Instances ……18
Table 4.9 table Instances ……18
Table 4.10 table Instances ……19
Table 4.11 table Instances ……19
Table 4.12 table Instances ……19
Table 4.13 table Instances ……19
Table 4.14 table Instances ……19
Table 4.15 table Instances ……19
Table 4.16 table Instances ……19
Table 4.17 table Instances ……19

1.Executive Summary

The Manufacturing Community is a student club at King Abdulaziz University, it was established in January 2017 by Dr. Ammar Mulaibari, it's a community that brings together innovators who have a passion for industry and entrepreneurial production. The club focuses on empowering innovators by helping them turn their ideas into tangible products and spreading a culture of making and manufacturing. Their vision is centralized on being the leading Student hub for manufacturing and innovation in the middle East.

The Club uses applications such as Excel, Google Sheets and Google Forms to collect and store data manually. The Club faces several problems during this process. First, errors in data entry are due to the absence of restrictions regulating how information must be entered into the designated fields. So, some users may enter invalid data. There are also limitations in sharing data directly between different departments, as each department keeps its data on a separate Excel sheet. In addition, there is difficulty in making modifications or updating data across all departments, as updates made in one department may not apply to the other department. Finally, using different applications to store data makes data transfer difficult and requires significant time and effort.

We obtained the necessary information to design a database system from the HR Manager at the Manufacturing Club, and this was achieved using two methods of data collection, which are interview and questionnaire. The interview was conducted via Zoom, the points that were discussed are: the Club's main activities and how they store and deal with the data. We also sent a questionnaire to the club's HR manager to find out about their business rules in data storage. The two data collection methods assisted us in understanding the issue facing the club and finding the right solution.

The club requires a centralized database system to manage and store data efficiently. The proposed system will include a unified database for all departments, enabling them to access and utilize data easily within a single system. The system also includes constraints to ensure the accuracy of data entered in each field in the required format and structure.

Additionally, the system will allow data to be modified and updated directly and simultaneously for all departments. Furthermore, the system will improve the quality of receiving and handling complaints and suggestions.

The primary goal of the database system is to offer centralized data management and easy access for multiple users. The database system ensures accuracy, consistency, as well as security of the data within an organization. In addition, it enables sharing of data within an organization while controlling access to sensitive information. Moreover, it enables easy retrieval as well as update of the data.

The database is significant for managing and organizing data in the club and have many benefits, the most important of which are Improving operational efficiency. Secure and organized storage of information and data, which reduces data damage and loss. improving redundancy planning, which makes information and data accurate. Supporting decision-making and this increases the reliability of information and data. Achieving integration between departments. This helps in managing information to improve decision making.

2.Conceptual Database Diagram Scenario

The manufacturing community club manages variety of educational events. Each event has unique ID, date, name and place. The event is presented by one or many instructors who have instructor ID, full name, phone number, e-mail and multiple social media accounts. An instructor may present one or more events.

Event must have at least one attendee that can attend one or many events. Each attendee has a SSN, phone number, name, Gender, and E-mail address.

The members may participate in many events either as participants or attendees. The members have Student ID, SSN, phone number, first and last name, department, position, multiple Email addresses, faculty, major, expected graduation year, gender assigned date and achievement such as awards, completed events, or certification. The achievement includes details such as number, date and description of the achievement.

The event is participated by at least 2 members and the role of each member is recorded with participation date.

Each member is assigned to only one department which includes department ID, department name, and multiple achievements of the department such as, achieving milestones, awards, or organizing events. Each achievement includes details such as number, date and description of the achievement. Each department must have at least one member.

A department also can organize zero or more events. Each event must be organized by at least one departments.

Each event must be sponsored by at least one sponsor that is have special ID, sponsor name, and phone number. The sponsor may or may not be sponsor for many events. The event must have multiple announcements that have announcement name, date and time. Many announcements are assigned for one event.

The complaints and suggestions are considered feedback. Each feedback contains feedback ID, feedback name, feedback type, and description. One or more feedback can be submitted by both members and attendees that may or may not send feedback.

2.1Entity – Relationship Diagram

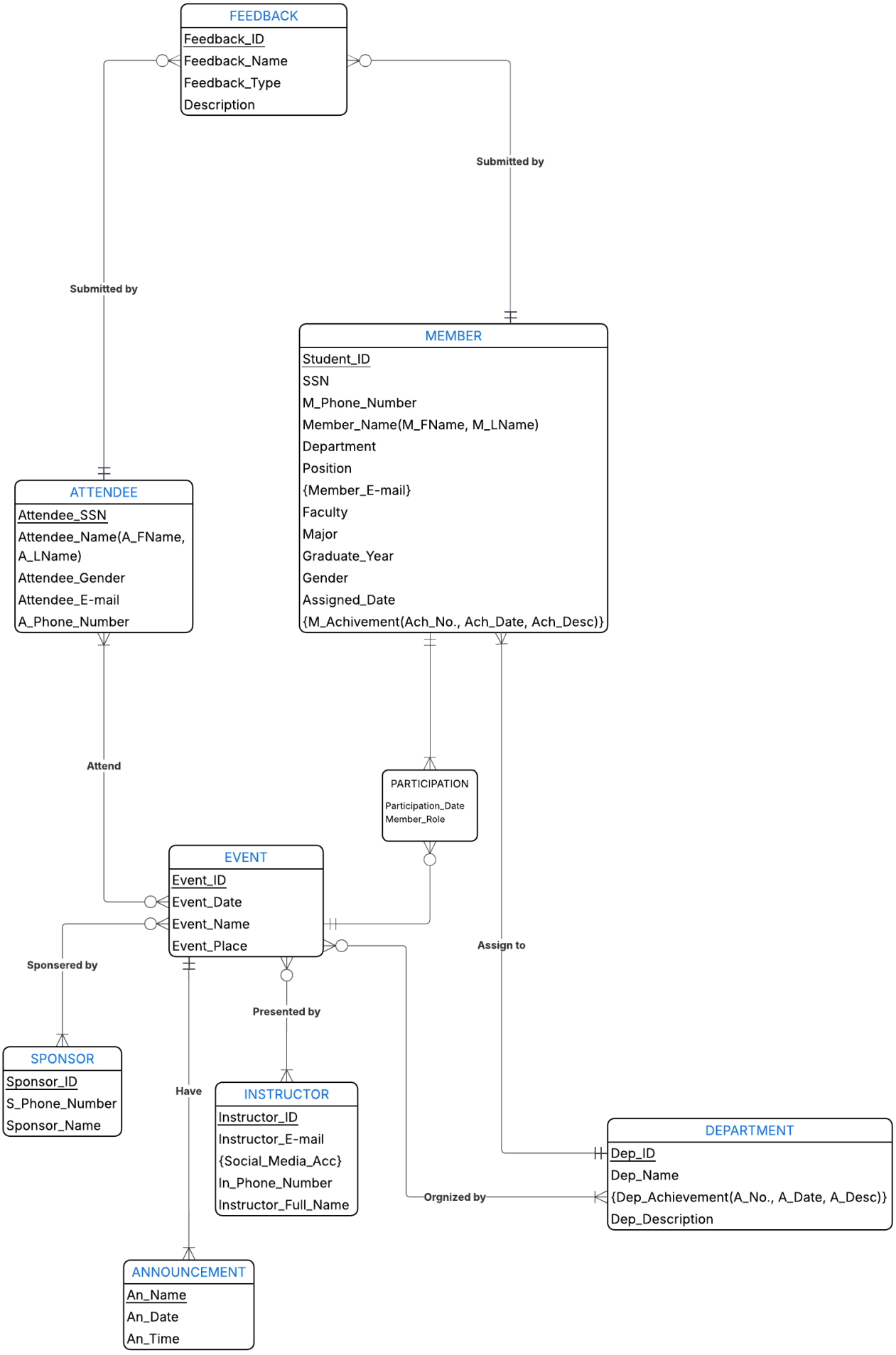


Figure 1.1 Entity Relationship Diagram (ERD)

2.2ERD Documentation

Entities

1.EVENT

Event (Event_ID, Event_Name, Event_Place, Event_Date)

2.MEMBER

Member (Student_ID, SSN, M_Phone_Number, Member_Name(M_FName, M_LName), Department, Position, {Member_E-mail} Faculty, Major, Graduate_Year, Gender, Assigned_Date, {M_Achivement(Ach_No., Ach_Date, Ach_Desc)})

3.DEPARTMENT

Department (Dep_ID, Dep_Name, Dep_Description, {M_Achivement(A_No., A_Date, A_Desc)})

4.INSTRUCTOR

Instructor (Instructor_ID, In_Phone_Number, Instructor_E-mail, {Social_Media_Acc}, Instructor_Full_Name)

5.SPONSOR

Sponsor (Sponsor_ID, S_Phone_Number, Sponsor_Name)

6.ANNONCEMENT

Announcement (An_Name, An_Date, An_Time)

7.FEEDBACK

Feedback (Feedback_ID, Feedback_Name, Feedback_Type, Description)

8.ATTENDEE

Attendee (Atendee_SSN, A_Phone_Number, Attendee_Name(A_FName, A_LName), Attendee_Gender, Attendee_E-mail)

Relationships

Event and Attendee

Event must have at least one attendee, and attendee may or may not attend many events, the relationship between the two entities **many-to-many.**

Event and Member

Event must be participated in by at least two members and the member may participate in zero or many events and the club keeps track of member role and their participation date , the relationship between the two entities **many-to-many.**

Event and Instructor

The event must be presented by at least one instructor , and the instructor may present one or more event, the relationship between the two entities **many-to-many.**

Event and sponsor

The event must be sponsored by at least one sponsor and the sponsor may or may not be sponsor for many event, the relationship between the two entities **many-to-many.**

Event and Announcement

The event must have multiple announcements, and the announcements must be assigned for one event. , the relationship between the two entities **one-to-many.**

Member and Department

The member is assigned to only one department, and the department must have at least one member, the relationship between the two entities **one-to-many.**

Department and Event

The department may organize zero or more events, and the event must be organized by at least three departments. **Many-to-many.**

Feedback and Member

The feedback is submitted by member that may or may not send one or more feedback, the relationship between the two entities **one-to-many.**

Feedback and Attendee

The feedback is submitted by attendee that may or may not send one or more feedback, the relationship between the two entities **one-to-many.**

3.Logical Database Design

At this stage, the ER diagram is converted into a logical data model (Relational Model). Where all attributes, primary and foreign keys are identified for each relation. Moreover, the Normalization stage is applied to check that all relation in the 3NF.

3.1Relational Model

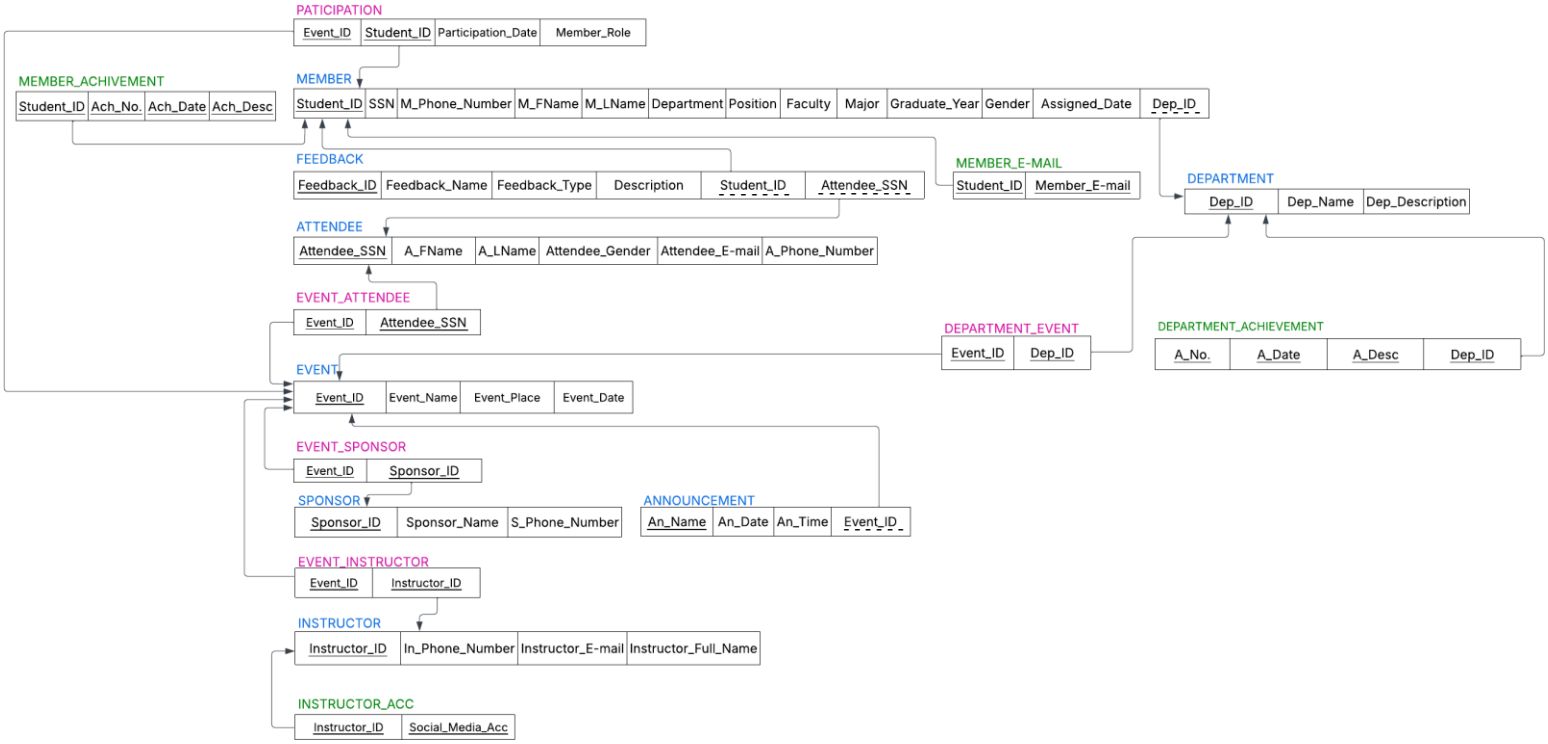


Figure 2.1 Relational Model Diagram

3.2Normalization

R1. PATICIPATION

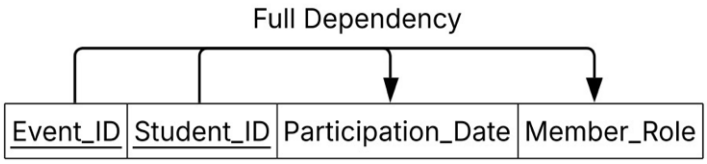


Figure 3.2.1 Normalization

FD1: Event_ID, Student_ID → Participation_Date, Member_Role

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.

Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.

Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R2. MEMBER

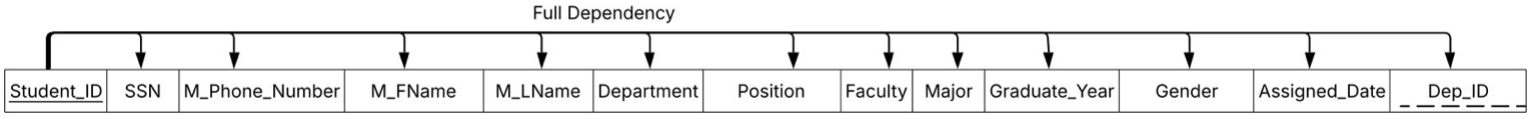


Figure 3.2.2 Normalization

FD1: Student_ID → SSN, M_Phone_Number, M_FName, M_LName, Department, Position, Faculty, Major, Graduate_Year, Gender, Assigned_Date, Dep_ID

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.

Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.

Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R3. MEMBER_ACHIVEMENT

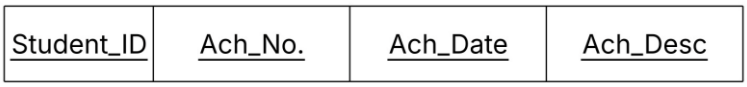


Figure 3.2.3 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.

Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.

Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R4. MEMBER_E-MAIL



Figure 3.2.4 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.

Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.

Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R5. FEEDBACK

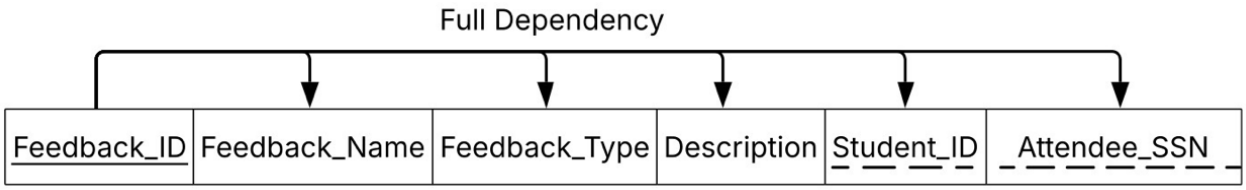


Figure 3.2.5 Normalization

FD1: Feedback _ID → Feedback_Name, Feedback_Type, Description, Student_ID, Attendee_SSN

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.

Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.

Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R6. ATTENDEE

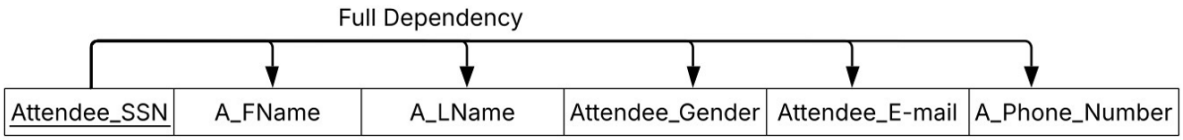


Figure 3.2.6 Normalization

FD1: Attendee_SSN → A_FName, A_LName, Attendee_Gender, Attendee_E-mail, A_Phone_Number

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R7. EVENT_ATTENDEE

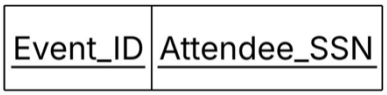


Figure 3.2.7 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R8. EVENT

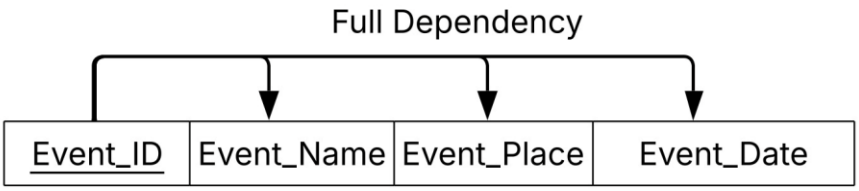


Figure 3.2.8 Normalization

FD1: Event _ID → Event_Name, Event_Place, Event_Date
First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R9. EVENT_SPONSOR

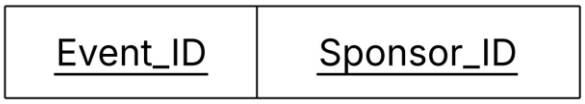


Figure 3.2.9 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R10. SPONSOR

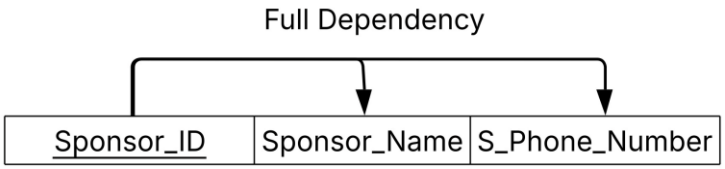


Figure 3.2.10 Normalization

FD1: Sponsor _ID → Sponsor_Name, S_Phone_Number

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R11. EVENT_INSTRUCTOR



Figure 3.2.11 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R12. INSTRUCTOR

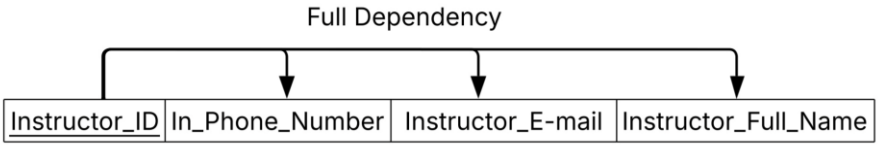


Figure 3.2.12 Normalization

FD1: Instructor _ID → In_Phone_Number, Instructor_E-mail, Instructor_Full_Name

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R13. INSTRUCTOR_ACC

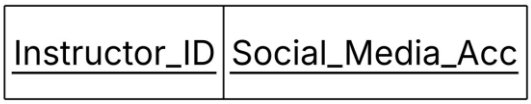


Figure 3.2.13 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R14. ANNOUNCEMENT

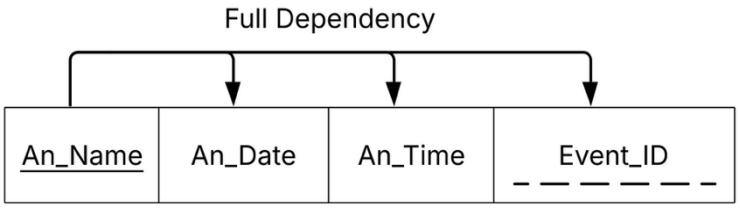


Figure 3.2.14 Normalization

FD1: An _Name → An_Date, An_Time, Event_ID
First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R15. DEPARTMENT

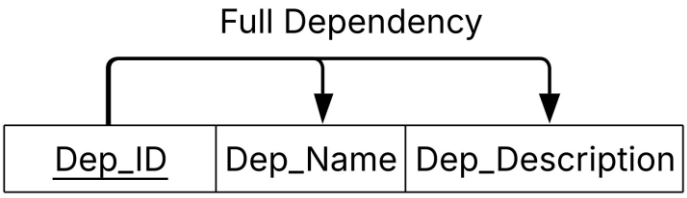


Figure 3.2.15 Normalization

FD1: Dep _ID → Dep_Name, Dep_Description
First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R16. DEPARTMENT_ACHIEVEMENT



Figure 3.2.16 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

R17. DEPARTMENT_EVENT

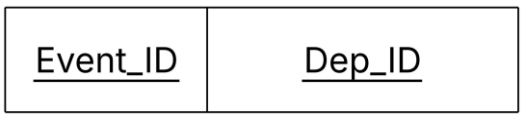


Figure 3.2.17 Normalization

First Normal Form:
All relation does not have multivalued attributes, so all in the 1NF.
Second Normal Form:
All relation does not have partial dependencies, so all in the 2NF.
Third Normal Form:
All relation does not have transitive dependencies, so all in the 3NF.

4.Database Implementation

In this step, we will create instance tables for each entity in the relational model, defining attributes, data types, primary keys, and foreign keys. This ensures a clear and consistent structure for implementing the database.

1. MEMBER

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data Type	Max Length
Student_ID	PK	NN,U			INT	
SSN		NN,U			VARCHAR	10
M_Phone_Number		NN			VARCHAR	10
M_FName		NN			VARCHAR	20
M_LName		NN			VARCHAR	20
Department		NN			VARCHAR	40
Position		NN			VARCHAR	30
Faculty		NN			VARCHAR	40
Major		NN			VARCHAR	40
Graduate_Year		NN			INT	
Gender		NN			VARCHAR	6
Assigned_Date		NN			DATE	
Dep_ID	FK	NN	DEPARTMENT	Dep_ID	INT	

Table 4.1 table Instances

2. DEPARTMENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data Type	Max Length
Dep_ID	PK	NN,U			INT	
Dep_Name		NN			VARCHAR	40
Dep_Description		NN			VARCHAR	100

Table 4.2 table Instances

3. EVENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data Type	Max Length
Event_ID	PK	NN,U			INT	
Event_Name		NN			VARCHAR	50
Event_Place		NN			VARCHAR	50
Event_Date		NN			DATE	

Table 4.3 table Instances

4. ATTENDEE

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Attendee_SSN	PK	NN, U			VARCHAR	10
A_FName		NN			VARCHAR	20
A_LName		NN			VARCHAR	20
Attendee_Gender		NN			VARCHAR	6
Attendee_Email		NN			VARCHAR	50
A_Phone_Number		NN/U			VARCHAR	10

Table 4.4 table Instances

5. FEEDBACK

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Feedback_ID	PK	NN, U			INT	
Feedback_Name		NN			VARCHAR	50
Feedback _Type		NN			VARCHAR	30
Description		NN			VARCHAR	350
Student_ID	FK	NN	MEMBER	Student_ID	INT	
Attendee_SSN	FK	NN	ATTENDEE	Attendee_SSN	VARCHAR	10

Table 4.5 table Instances

6. PARTICIPATION

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Event_ID	PK, FK	NN, U	EVENT	Event_ID	INT	
Student_ID	PK, FK	NN, U	MEMBER	Student_ID	INT	
Participation_Date		NN			DATE	
Member_Role		NN			VARCHAR	50

Table 4.6 table Instances

7. SPONSOR

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Sponsor_ID	PK	NN, U			INT	
Sponsor_Name		NN			VARCHAR	50
S_Phone_Number		NN			VARCHAR	10

Table 4.7 table Instances

8. EVENT_SPONSOR

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Event_ID	PK, FK	NN, U	EVENT	Event_ID	INT	
Sponsor_ID	PK, FK	NN, U	SPONSOR	Sponsor_ID	INT	

Table 4.8 table Instances

9.EVENT_ATTENDEE

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Event_ID	PK, FK	NN, U	EVENT	Event_ID	INT	
Attendee_SSN	PK, FK	NN, U	ATTENDEE	Attendee_SSN	VARCHAR	10

Table 4.9 table Instance

10. INSTRUCTOR

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Instructor_ID	PK	NN, U			INT	
In_Phone_Number		NN			VARCHAR	10
Instructor_Email		NN			VARCHAR	50
Instructor_Full_Name		NN			VARCHAR	60

Table 4.10 table Instances

11. EVENT_INSTRUCTOR

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Event_ID	PK, FK	NN, U	EVENT	Event_ID	INT	
Instructor_ID	PK, FK	NN, U	INSTRUCTOR	Instructor_ID	INT	

Table 4.11 table Instances

12. INSTRUCTOR_ACC

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Instructor_ID	PK, FK	NN, U	INSTRUCTOR	Instructor_ID	INT	
Social_Media_Acc	PK	NN, U			VARCHAR	150

Table 4.12 table Instances

13. ANNOUNCEMENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
An_Name	PK	NN, U			VARCHAR	100
An_Date		NN			DATE	
An_Time		NN			TIME	
Event_ID	FK	NN	EVENT	Event_ID	INT	

Table 4.13 table Instances

14. MEMBER_EMAIL

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Student_ID	PK, FK	NN	MEMBER	Student_ID	INT	
Member_Email	PK	NN,U			VARCHAR	50

Table 4.14 table Instances

15. MEMBER_ACHIEVEMENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Student_ID	PK, FK	NN	MEMBER	Student_ID	INT	
Ach_No	PK	NN/U			INT	
Ach_Date	PK	NN/U			DATE	
Ach_Desc	PK	NN/U			VARCHAR	150

Table 4.15 table Instances

16. DEPARTMENT_EVENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
Event_ID	PK, FK	NN, U	EVENT	Event_ID	INT	
Dep_ID	PK, FK	NN, U	DEPARTMENT	Dep_ID	INT	

Table 4.16 table Instances

17. DEPARTMENT_ACHIEVEMENT

Column Name	Key Type	Null/Unique	FK Table	FK Column	Data type	Max length
A_No	PK	NN, U			INT	
A_Date	PK	NN, U			DATE	
A_Desc	Pk	NN, U			VARCHAR	150
Dep_ID	PK, FK	NN	DEPARTMENT	Dep_ID	INT	

Table 4.17 table Instances

5.Database Implementation in MySQL

In this section, a database was implemented using MySQL. This includes creating tables with their primary and foreign keys, inserting sample data, and applying necessary constraints and modifications.

5.1 Tables creation

```
-- Create MEMBER table
CREATE TABLE MEMBER (
Student_ID INT NOT NULL,
SSN VARCHAR (10) NOT NULL,
M_Phone_Number VARCHAR (10) NOT NULL,
M_FName VARCHAR (20) NOT NULL,
M_LName VARCHAR (20) NOT NULL,
Department VARCHAR (40) NOT NULL,
Position VARCHAR (30) NOT NULL,
Faculty VARCHAR (40) NOT NULL,
Major VARCHAR (40) NOT NULL,
Graduate_Year INT NOT NULL,
Gender VARCHAR (6) NOT NULL,
Assigned_Date DATE NOT NULL,
Dep_ID INT,
CONSTRAINT MEMBER_PK PRIMARY KEY (Student_ID),
CONSTRAINT MEMBER_SSN_UK UNIQUE (SSN),
CONSTRAINT MEMBER_DEPT_FK FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT
(Dep_ID)
ON DELETE SET NULL
);
```

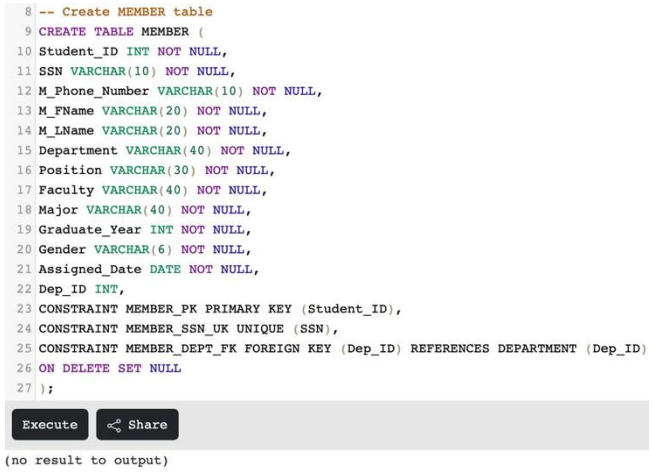


Figure 5.1-1: Tables creation

```
-- Create DEPARTMENT table
CREATE TABLE DEPARTMENT (
Dep_ID INT NOT NULL,
Dep_Name VARCHAR (40) NOT NULL,
Dep_Description VARCHAR (100) NOT NULL,
CONSTRAINT DEPARTMENT_PK PRIMARY KEY (Dep_ID)
);
```



Figure 5.1-2: Tables creation

```
-- Create EVENT table

CREATE TABLE EVENT (
Event_ID INT NOT NULL,
Event_Name VARCHAR (50) NOT NULL,
Event_Place VARCHAR (50) NOT NULL,
Event_Date DATE NOT NULL,
CONSTRAINT EVENT_PK PRIMARY KEY (Event_ID)
);
```

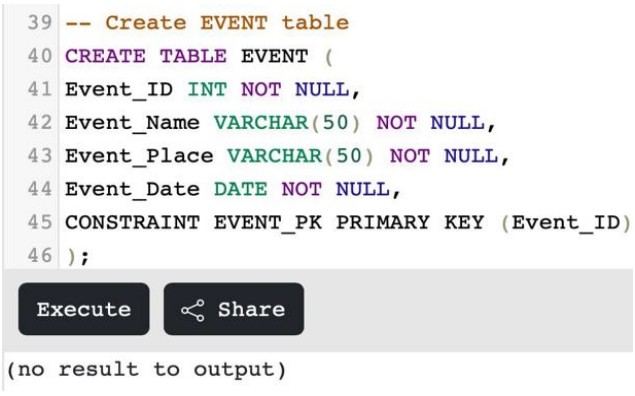


Figure 5.1-3: Tables creation

```
-- Create ATTENDEE table

CREATE TABLE ATTENDEE (
Attendee_SSN VARCHAR (15) NOT NULL,
A_FName VARCHAR (20) NOT NULL,
A_LName VARCHAR (20) NOT NULL,
Attendee_Gender VARCHAR (10) NOT NULL,
Attendee_Email VARCHAR (50) NOT NULL,
A_Phone_Number VARCHAR (10) NOT NULL,
CONSTRAINT ATTENDEE_PK PRIMARY KEY (Attendee_SSN),
CONSTRAINT ATTENDEE_PHONE_UK UNIQUE (A_Phone_Number));
```

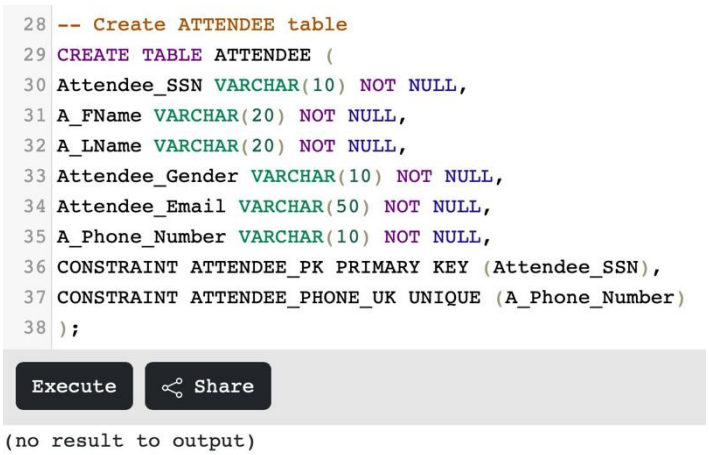


Figure 5.1-4: Tables creation

```
-- Create FEEDBACK table
CREATE TABLE FEEDBACK (
Feedback_ID INT NOT NULL,
Feedback_Name VARCHAR (50) NOT NULL,
Feedback_Type VARCHAR (30) NOT NULL,
Description VARCHAR (350) NOT NULL,
Student_ID INT,
Attendee_SSN VARCHAR (10),
CONSTRAINT FEEDBACK_PK PRIMARY KEY (Feedback_ID),
CONSTRAINT FEEDBACK_MEMBER_FK FOREIGN KEY (Student_ID) REFERENCES MEMBER
(Student_ID)
ON DELETE SET NULL,
CONSTRAINT FEEDBACK_ATTENDEE_FK FOREIGN KEY (Attendee_SSN) REFERENCES
ATTENDEE (Attendee_SSN)
ON DELETE SET NULL
);
```

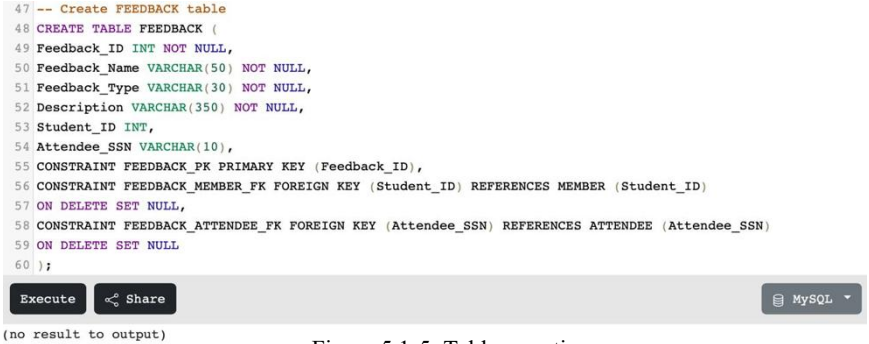


Figure 5.1-5: Tables creation

```
-- Create PARTICIPATION table
CREATE TABLE PARTICIPATION (
Event_ID INT NOT NULL,
Student_ID INT NOT NULL,
Participation_Date DATE NOT NULL,
Member_Role VARCHAR (50) NOT NULL,
CONSTRAINT PARTICIPATION_PK PRIMARY KEY (Event_ID, Student_ID),
CONSTRAINT PARTICIPATION_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES
EVENT (Event_ID)
ON DELETE CASCADE,
CONSTRAINT PARTICIPATION_MEMBER_FK FOREIGN KEY (Student_ID) REFERENCES
MEMBER (Student_ID)
ON DELETE CASCADE
);
```

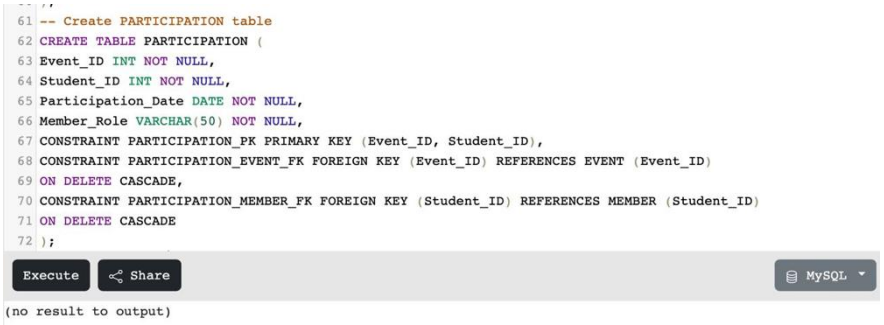


Figure 5.1-6: Tables creation

```
-- Create SPONSOR table
CREATE TABLE SPONSOR (
Sponsor_ID INT NOT NULL,
Sponsor_Name VARCHAR (50) NOT NULL,
S_Phone_Number VARCHAR (10) NOT NULL,
CONSTRAINT SPONSOR_PK PRIMARY KEY (Sponsor_ID)
);
```

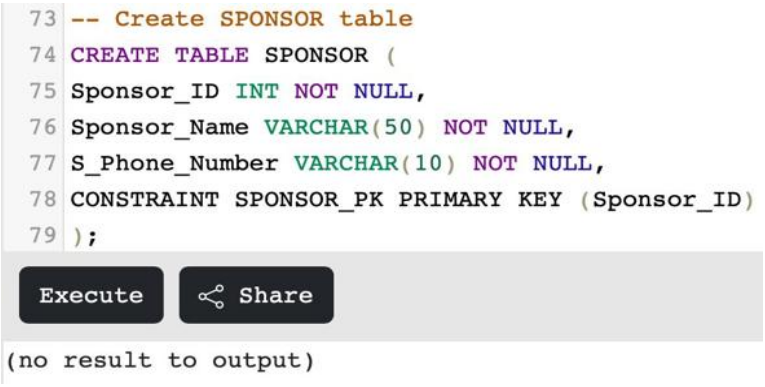


Figure 5.1-7: Tables creation

```
-- Create EVENT_SPONSOR table
CREATE TABLE EVENT_SPONSOR (
Event_ID INT NOT NULL,
Sponsor_ID INT NOT NULL,
CONSTRAINT EVENT_SPONSOR_PK PRIMARY KEY (Event_ID, Sponsor_ID),
```

```
CONSTRAINT EVENT_SPONSOR_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES
EVENT (Event_ID)
ON DELETE CASCADE,
CONSTRAINT EVENT_SPONSOR_SPONSOR_FK FOREIGN KEY (Sponsor_ID) REFERENCES
SPONSOR (Sponsor_ID)
ON DELETE CASCADE
);
```

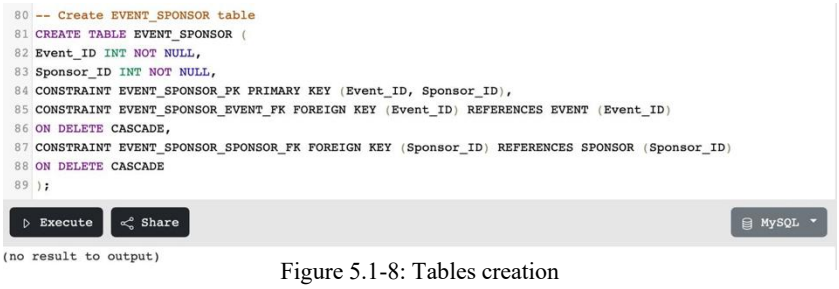


Figure 5.1-8: Tables creation

```
-- Create EVENT_ATTENDEE table
CREATE TABLE EVENT_ATTENDEE (
Event_ID INT NOT NULL,
Attendee_SSN VARCHAR (10) NOT NULL,
CONSTRAINT EVENT_ATTENDEE_PK PRIMARY KEY (Event_ID, Attendee_SSN),
CONSTRAINT EVENT_ATTENDEE_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES
EVENT (Event_ID)
ON DELETE CASCADE,
CONSTRAINT EVENT_ATTENDEE_ATTENDEE_FK FOREIGN KEY (Attendee_SSN)
REFERENCES ATTENDEE (Attendee_SSN)
ON DELETE CASCADE
);
```

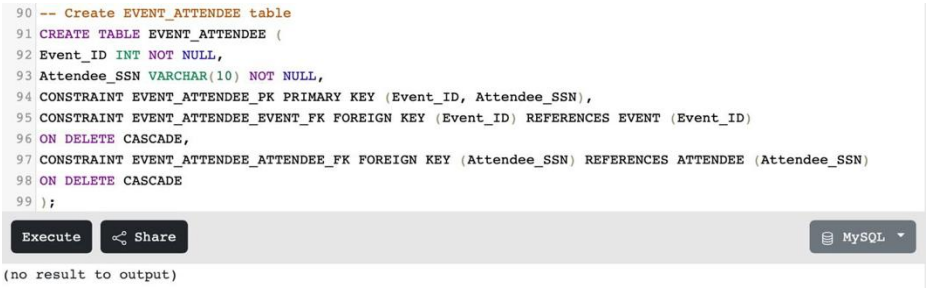


Figure 5.1-9: Tables creation

```
-- Create INSTRUCTOR table
CREATE TABLE INSTRUCTOR (
Instructor_ID INT NOT NULL,
Instructor_Full_Name VARCHAR (60) NOT NULL,
In_Phone_Number VARCHAR (10) NOT NULL,
Instructor_Email VARCHAR (50) NOT NULL,
CONSTRAINT INSTRUCTOR_PK PRIMARY KEY (Instructor_ID)
);
```

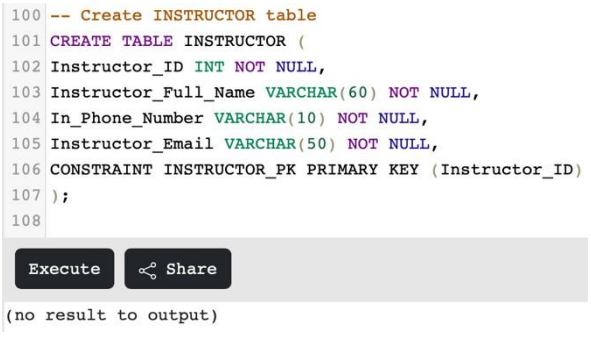


Figure 5.1-10: Tables creation

```
-- Create EVENT_INSTRUCTOR table
CREATE TABLE EVENT_INSTRUCTOR (
Event_ID INT NOT NULL,
Instructor_ID INT NOT NULL,
CONSTRAINT EVENT_INSTRUCTOR_PK PRIMARY KEY (Event_ID, Instructor_ID),
CONSTRAINT EVENT_INST_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES EVENT
(Event_ID)
ON DELETE CASCADE,
CONSTRAINT EVENT_INST_INSTRUCTOR_FK FOREIGN KEY (Instructor_ID)
REFERENCES INSTRUCTOR (Instructor_ID)
ON DELETE CASCADE
;(
```

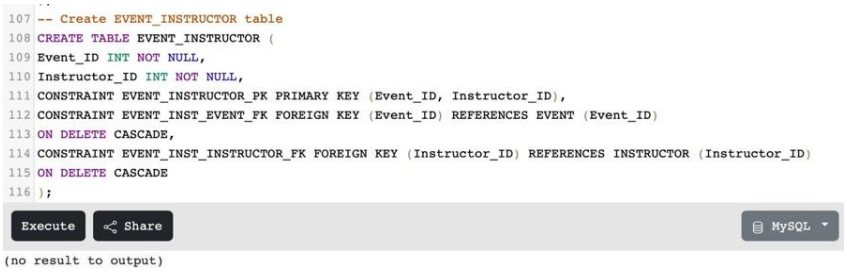


Figure 5.1-11: Tables creation

```
-- Create INSTRUCTOR_ACC table
CREATE TABLE INSTRUCTOR_ACC (
Instructor_ID INT NOT NULL,
Social_Media_Acc VARCHAR (150) NOT NULL,
CONSTRAINT INSTRUCTOR_ACC_PK PRIMARY KEY (Instructor_ID,
Social_Media_Acc),
CONSTRAINT INSTRUCTOR_ACC_FK FOREIGN KEY (Instructor_ID) REFERENCES
INSTRUCTOR (Instructor_ID)
ON DELETE CASCADE
);
```

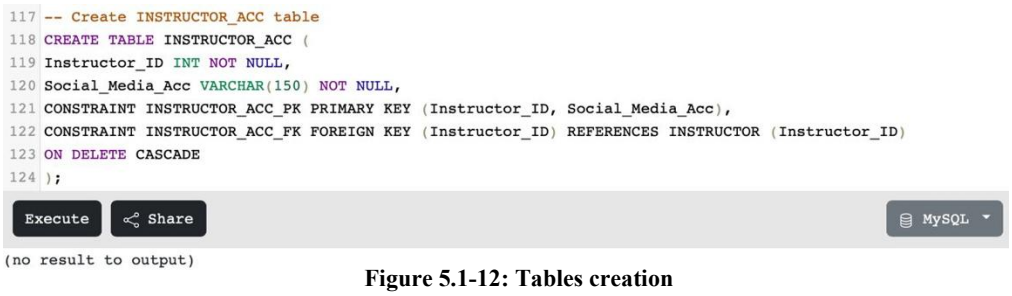


Figure 5.1-12: Tables creation

```
-- Create ANNOUNCEMENT table
CREATE TABLE ANNOUNCEMENT (
An_Name VARCHAR (100) NOT NULL,
An_Date DATE NOT NULL,
An_Time TIME NOT NULL,
Event_ID INT,
CONSTRAINT ANNOUNCEMENT_PK PRIMARY KEY (An_Name),
```

```
CONSTRAINT ANNOUNCEMENT_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES EVENT
(Event_ID)
ON DELETE SET NULL
);
```

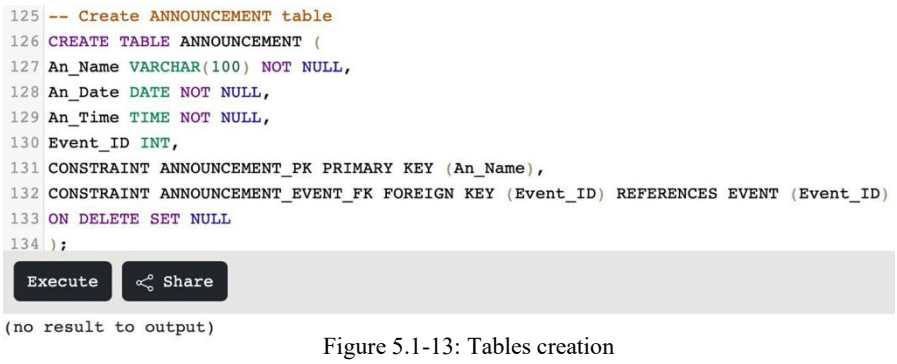


Figure 5.1-13: Tables creation

```
-- Create MEMBER_EMAIL table
CREATE TABLE MEMBER_EMAIL (
Student_ID INT NOT NULL,
Member_Email VARCHAR (50) NOT NULL,
CONSTRAINT MEMBER_EMAIL_PK PRIMARY KEY (Student_ID, Member_Email),
CONSTRAINT MEMBER_EMAIL_FK FOREIGN KEY (Student_ID) REFERENCES MEMBER
(Student_ID)
ON DELETE CASCADE
);
```



Figure 5.1-14: Tables creation

```
-- Create MEMBER_ACHIEVEMENT table
CREATE TABLE MEMBER_ACHIEVEMENT (
Student_ID INT NOT NULL,
Ach_No INT NOT NULL,
Ach_Date DATE NOT NULL,
Ach_Desc VARCHAR (150) NOT NULL,
CONSTRAINT MEMBER_ACHIEVEMENT_PK PRIMARY KEY (Student_ID, Ach_No,
Ach_Date, Ach_Desc),
CONSTRAINT MEMBER_ACHIEVE_MEMBER_FK FOREIGN KEY (Student_ID) REFERENCES
MEMBER (Student_ID)
ON DELETE CASCADE
);
```

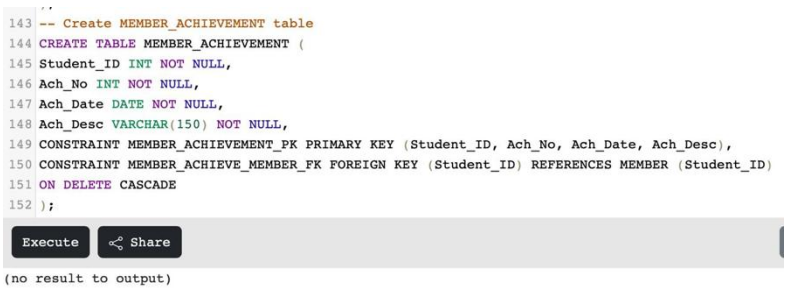


Figure 5.1-15: Tables creation

```
-- Create DEPARTMENT_EVENT table
CREATE TABLE DEPARTMENT_EVENT (
Event_ID INT NOT NULL,
Dep_ID INT NOT NULL,
CONSTRAINT DEPARTMENT_EVENT_PK PRIMARY KEY (Event_ID, Dep_ID),
CONSTRAINT DEPT_EVENT_EVENT_FK FOREIGN KEY (Event_ID) REFERENCES EVENT
(Event_ID)
ON DELETE CASCADE,
CONSTRAINT DEPT_EVENT_DEPT_FK FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT
(Dep_ID)
ON DELETE CASCADE
);
```

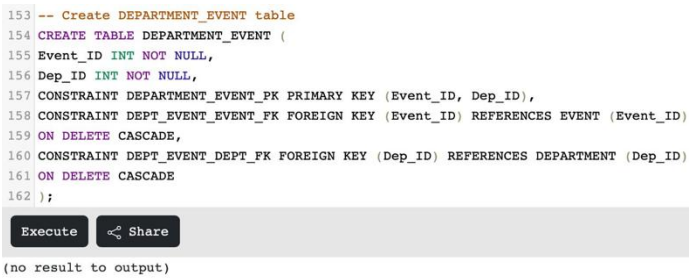


Figure 5.1-16: Tables creation

```
-- Create DEPARTMENT_ACHIEVEMENT table
CREATE TABLE DEPARTMENT_ACHIEVEMENT (
A_No INT NOT NULL,
A_Date DATE NOT NULL,
A_Desc VARCHAR (150) NOT NULL,
Dep_ID INT NOT NULL,
CONSTRAINT DEPT_ACHIEVE_PK PRIMARY KEY (A_No, A_Date, A_Desc, Dep_ID),
CONSTRAINT DEPT_ACHIEVE_FK FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT
(Dep_ID)
ON DELETE CASCADE
);
```

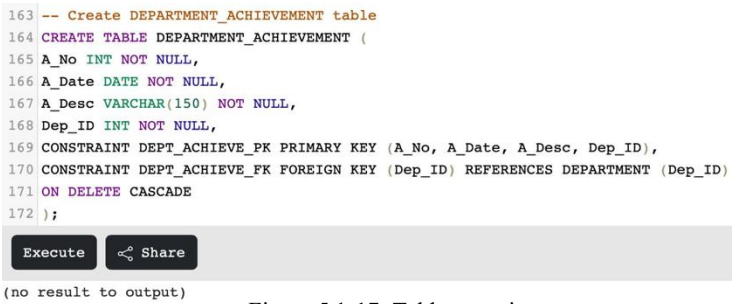


Figure 5.1-17: Tables creation

5.2 Insert into Tables

```
-- Insert into DEPARTMENT Table
INSERT INTO DEPARTMENT (Dep_ID, Dep_Name, Dep_Description) VALUES
(1,'Human Resources','Responsible for internal members and organizing
member affairs'),
(2,'Operation','Responsible for planning and managing events inside and
outside the club'),
(3,'Marketing','Responsible for advertisements and the external image of
the club'),
(4,'Financial Management','Responsible for the club budget, expenses,
income, and invoice management'),
(5,'Public Relations','Responsible for communicating with sponsors and
speakers and searching for event venues');
```

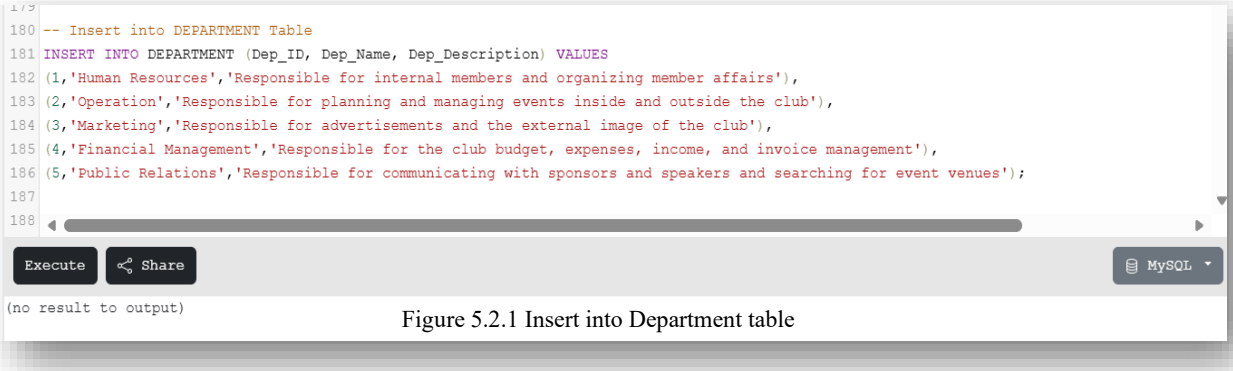


Figure 5.2.1 Insert into Department table

```
-- Insert into MEMBER Table
INSERT INTO MEMBER
(Student_ID, SSN, M_Phone_Number, M_FName, M_LName, Department,
Position, Faculty, Major, Graduate_Year, Gender, Assigned_Date, Dep_ID)
VALUES
(2300011, '1234567890', '0501112223', 'Ahmed', 'Khaled', 'Operation',
'Member', 'Since', 'Math', 2026, 'Male', '2024-09-01', 2),
(2300012, '2345678901', '0501112224', 'Sara', 'Omar', 'Human Resources',
'Member', 'Engineering', 'Electrical Engineering', 2027, 'Female',
'2024-09-01', 1),
(2300013, '3456789012', '0501112225', 'Yousef', 'Ali', 'Marketing',
'Member', 'Engineering', 'Computer Engineering', 2025, 'Male', '2024-09-
01', 3),
(2300014,'1029384760','0501234570','Noura','Hassan','Public
Relations','Designer','Computing','Computer
Science',2028,'Female','2025-01-18',5),
(2200015,'1029384761','0501234571','Yousef','Khaled','Operation','Conten
t Writer','Business','Marketing',2027,'Male','2025-01-20',2);
```

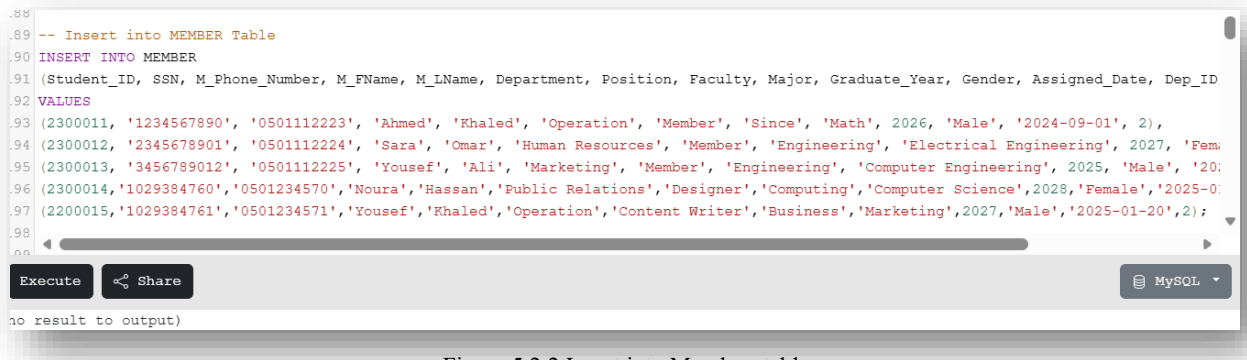


Figure 5.2.2 Insert into Member table

```
-- Insert into ATTENDEE Table
INSERT INTO ATTENDEE
(Attendee_SSN, A_FName, A_LName, Attendee_Gender, Attendee_Email,
A_Phone_Number)
VALUES
('1276543210', 'Noura', 'Hassan', 'Female', 'noura@gmail.com',
'0552003001'),
('1165432109', 'Khalil', 'Mansour', 'Male', 'khalil@gmail.com',
'0552003002'),
('2011223346','Dana','Ahmed','Female','dana@gmail.com','0551112235'),
('2011223348','Reem','Hadi','Female','reem@gmail.com','0551112237'),
('2011223347','Abdulrahman','Saad','Male','abdulrahman@gmail.com','05511
12236');
```

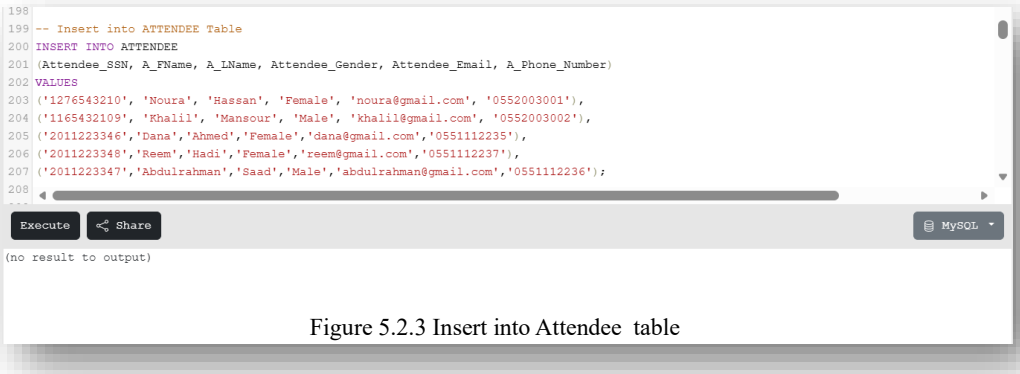


Figure 5.2.3 Insert into Attendee table

```
-- Insert into EVENT Table
INSERT INTO EVENT (Event_ID, Event_Name, Event_Place, Event_Date) VALUES
(101,'AI Workshop','Faculty of Computing Hall','2026-05-10'),
(102,'Cybersecurity Bootcamp','Building 31','2026-06-20'),
(103,'Industrial Engineering Seminar','King Faisal Conference
Center','2026-06-15'),
(104, 'Manufacturing Basics Workshop', 'Campus - Hall A', '2026-05-20'),
(105, 'Innovation & Entrepreneurship Night', 'Innovation Center', '2026-
06-10');
```

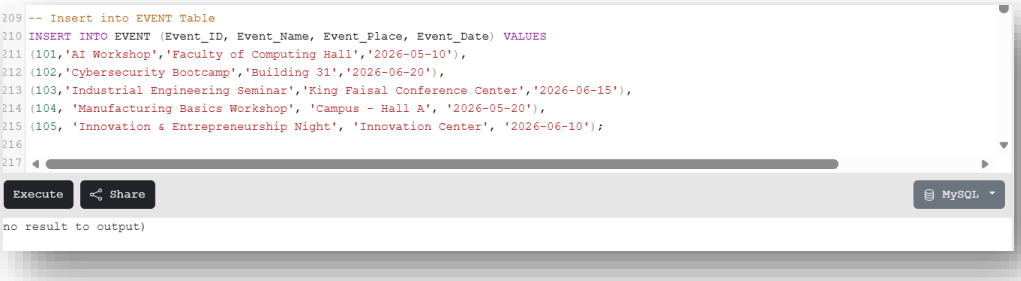


Figure 5.2.4 Insert into Event table

```
-- Insert into PARTICIPATION Table
INSERT INTO PARTICIPATION (Event_ID, Student_ID, Participation_Date,
Member_Role) VALUES
(104, 2300011, '2026-05-20', 'Orgnaizer'),
(104, 2300014,  DEFAULT    , 'Volunteer'),
(105, 2300011, '2026-06-10', 'Volunteer'),
(103, 2300013, '2026-06-15', 'Photographer'),
(103, 2300012, '2026-06-15', 'Volunteer'),
(102, 2300011, '2026-06-20', 'Orgnaizer'),
(101, 2200015, '2026-05-10', 'Volunteer'),
(101, 2300012, '2026-05-10', 'Photographer');
```

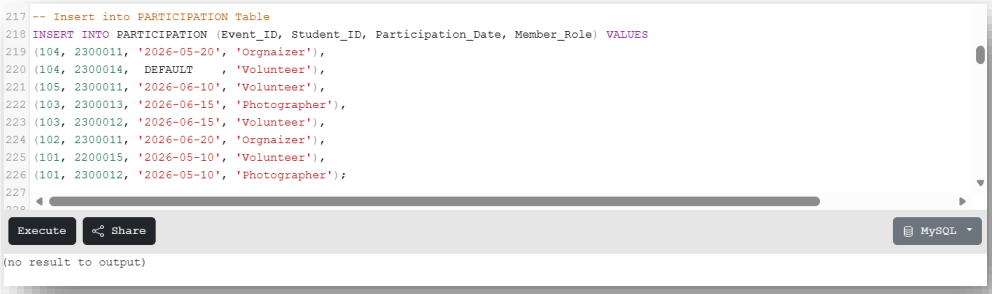


Figure 5.2.5 Insert into Participation table

```
-- Insert into EVENT_ATTENDEE Table
INSERT INTO EVENT_ATTENDEE (Event_ID, Attendee_SSN) VALUES
(101, '2011223346'),
(102, '2011223348'),
(103, '2011223347'),
(103, '1165432109'),
(104, '2011223346'),
(105, '1276543210'),
(105, '1165432109'),
(105, '2011223346');
```



Figure 5.2.6 Insert into Event-Attendee table

```
-- Insert into INSTRUCTOR Table
INSERT INTO INSTRUCTOR (Instructor_ID, In_Phone_Number,
Instructor_Email, Instructor_Full_Name) VALUES
(3001,'0502001111','dr.ahmad@kau.edu.sa', 'Dr. Ahmad Al-Shehri'),
(3002,'0502002222','sara@kau.edu.sa', 'Sara Al-Qahtani'),
(3003,'0502003333','eng.khaled@kau.edu.sa', 'Eng. Khaled Al-Harbi'),
(3004,'0502004444','dr.noura@kau.edu.sa', 'Dr. Noura Al-Zahrani'),
(3005,'0502005555','faisal@kau.edu.sa', 'Faisal Al-Otaibi');
```

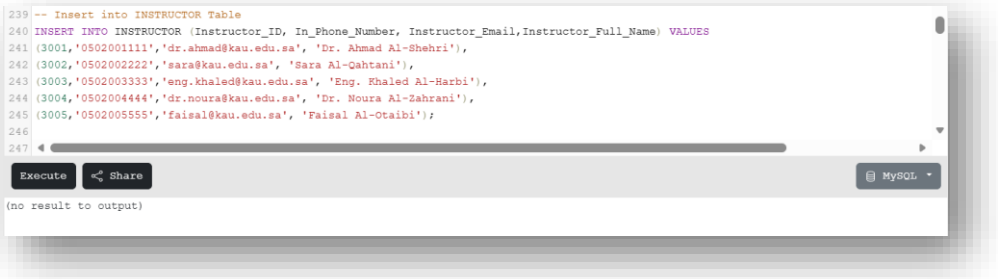


Figure 5.2.7 Insert into Instructor table

```
-- Insert into EVENT_INSTRUCTOR Table
INSERT INTO EVENT_INSTRUCTOR (Event_ID, Instructor_ID) VALUES
(101, 3005),
(101, 3003),
(102, 3004),
(102, 3003),
(103, 3001),
(103, 3002),
(104, 3002),
(104, 3001);
```



Figure 5.2.8 Insert into Event-Instructor table

```
-- Insert into INSTRUCTOR_ACC Table
INSERT INTO INSTRUCTOR_ACC (Instructor_ID, Social_Media_Acc) VALUES
(3001, 'https://twitter.com/inst1'),
(3001, 'https://linkedin.com/in/inst1'),
(3002, 'https://twitter.com/inst2'),
(3003, 'https://twitter.com/inst3'),
(3003, 'https://linkedin.com/in/inst3'),
(3004, 'https://twitter.com/inst4'),
(3004, 'https://instagram.com/inst4'),
(3005, 'https://twitter.com/inst5');
```



Figure 5.2.9 Insert into Instructor _ACC table

```
-- Insert into SPONSOR Table
INSERT INTO SPONSOR (Sponsor_ID, Sponsor_Name, S_Phone_Number) VALUES
(4001, 'TechNova', '0558009001'),
(4002, 'BuildPro', '0558009002'),
(4003, 'SmartSolutions', '0558009003'),
(4004, 'FutureTech', '0558009004'),
(4005, 'EduPartners', '0558009005');
```



Figure 5.2.10 Insert into Sponsor table

```
-- Insert into EVENT_SPONSOR Table
INSERT INTO EVENT_SPONSOR (Event_ID, Sponsor_ID) VALUES
(101, 4001),
(101, 4002),
(102, 4001),
(102, 4003),
(103, 4002),
(103, 4004),
(104, 4003),
(105, 4005);
```

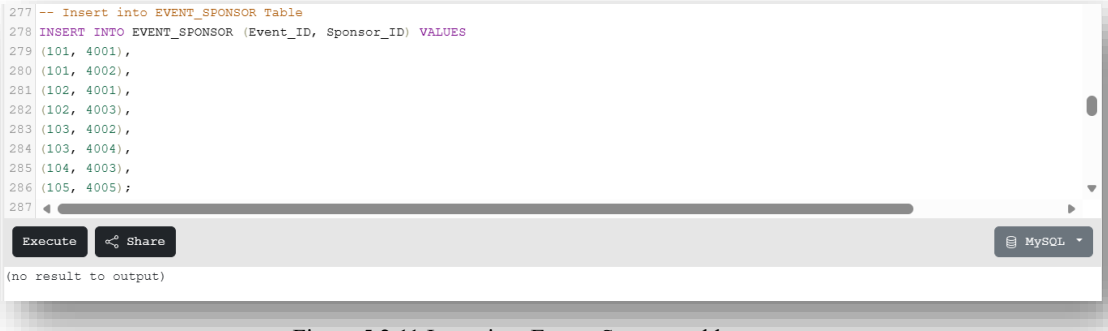


Figure 5.2.11 Insert into Event- Sponsor table

```
-- Insert into ANNOUNCEMENT Table
INSERT INTO ANNOUNCEMENT (An_Name, An_Date, An_Time, Event_ID) VALUES
('Workshop Opening', '2026-05-18', '18:00:00', 101),
('Registration Reminder', '2026-05-19', '20:00:00', 101),
('Keynote Start', '2026-06-08', '19:30:00', 102),
('Cybersecurity Update', '2026-05-21', '17:00:00', 103),
('Leadership Event Announcement', '2026-05-24', '16:30:00', 104);
```

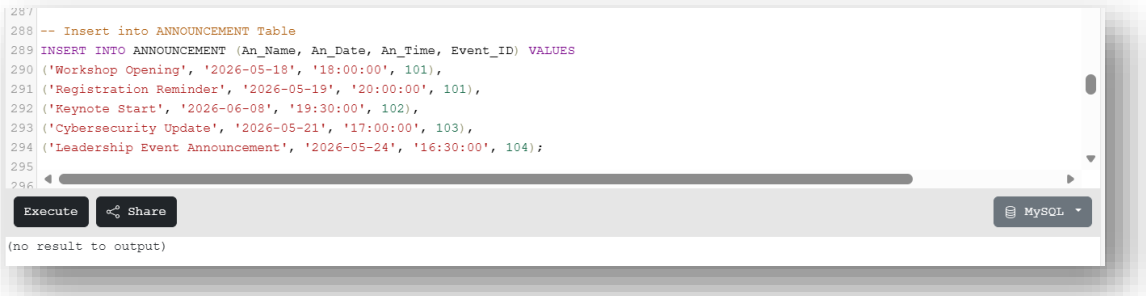


Figure 5.2.12 Insert into Announcement table

```
-- Insert into MEMBER_EMAIL Table
INSERT INTO MEMBER_EMAIL (Student_ID, Member_Email) VALUES
(2300011, 'ahmed.kh@gmail.com'),
(2300011, 'ahmed.m@kau.edu.sa'),
(2300012, 'sara.omar@kau.edu.sa'),
(2300013, 'yousef.ali@yahoo.com'),
(2300013, 'yousef.ali@kau.edu.sa'),
(2300014, 'noura.hassan@gmail.com'),
(2300014, 'noura.hassan@kau.edu.sa'),
(2200015, 'yousef.khaled@gmail.com');
```



Figure 5.2.13 Insert into Member-Email table

```
-- Insert into MEMBER_ACHIEVEMENT Table
INSERT INTO MEMBER_ACHIEVEMENT (Student_ID, Ach_No, Ach_Date, Ach_Desc)
VALUES
(2300011, 1, '2026-05-10', 'Best Organizer Award'),
(2300011, 2, '2026-05-15', 'Outstanding Participation'),
(2300012, 1, '2026-05-12', 'Excellent Team Work'),
(2300011, 3, '2026-05-18', 'Top Speaker Award'),
(2300013, 1, '2026-05-20', 'Event Contribution Recognition'),
(2300014, 1, '2026-05-22', 'Design Excellence Award'),
(2200015, 1, '2026-05-25', 'Leadership Recognition'),
(2200015, 2, '2026-05-28', 'Active Member Award');
```

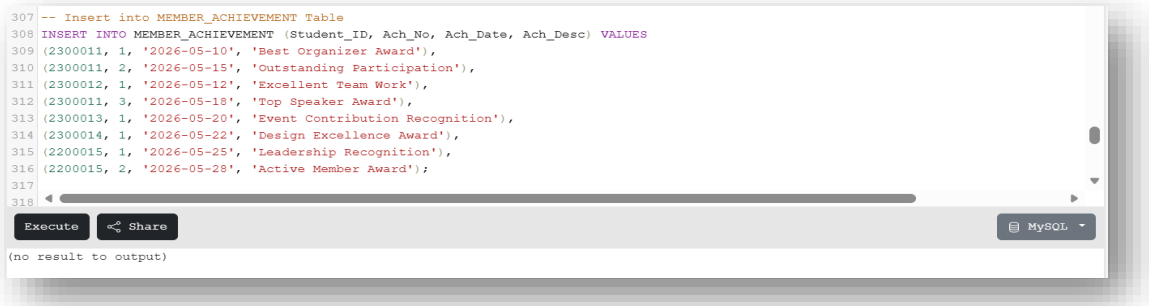


Figure 5.2.14 Insert into Member-Achievement table

```
-- Insert into DEPARTMENT_EVENT Table
INSERT INTO DEPARTMENT_EVENT (Event_ID, Dep_ID) VALUES
(101, 1),
(101, 2),
(102, 2),
(102, 3),
(103, 3),
(103, 4),
(104, 4),
(105, 2);
```

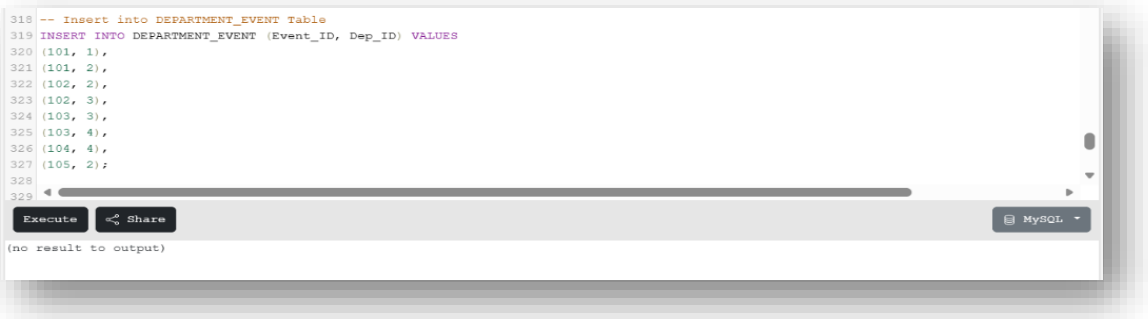


Figure 5.2.15 Insert into Department-Event table

```
-- Insert into DEPARTMENT_ACHIEVEMENT Table
INSERT INTO DEPARTMENT_ACHIEVEMENT (A_No, A_Date, A_Desc, Dep_ID) VALUES
(1, '2026-05-10', 'Best Event Organization Award', 1),
(1, '2026-05-10', 'Best Event Organization Award', 2),
(2, '2026-05-12', 'Outstanding HR Performance', 2),
(1, '2026-05-12', 'Outstanding HR Performance', 3),
(2, '2026-05-15', 'Technical Excellence in Events', 3),
(3, '2026-05-15', 'Technical Excellence in Events', 4),
(5, '2026-05-18', 'Best Event Execution Award', 4),
(1, '2026-05-20', 'Top Communication and Outreach', 5);
```

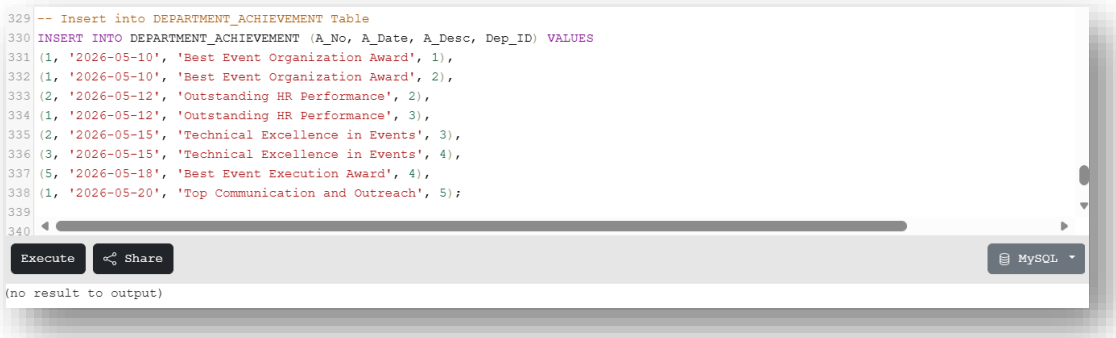


Figure 5.2.16 Insert into Department-Achievement table

```
-- Insert into FEEDBACK Table
INSERT INTO FEEDBACK (Feedback_ID, Feedback_Name, Feedback_Type,
Description, Student_ID, Attendee_SSN) VALUES
(1, 'Event Organization Issue', 'Complaint', 'Time delay in starting the
event', 2300011, NULL),
(2, 'Great Experience', 'Positive', 'Very well organized and informative
event', NULL, '2011223347'),
(3, 'Suggestion for Improvement', 'Suggestion', 'Need more seating areas
in events', NULL, '1276543210'),
(4, 'Technical Problem', 'Complaint', 'Sound system was not clear during
workshop', 2300013, NULL),
(5, 'Excellent Event', 'Positive', 'Loved the cybersecurity workshop
content', NULL, '2011223348');
```

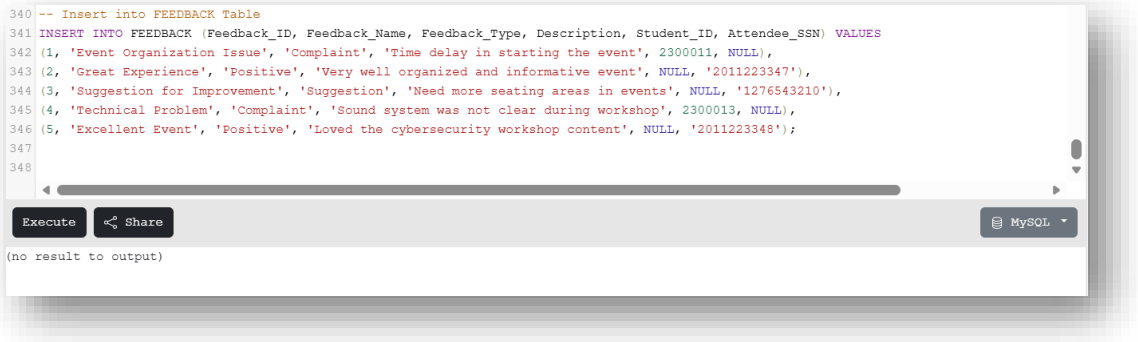


Figure 5.2.17 Insert into Feedback table

5.3 Table Modification

```
-- Applying a modification to add a default value
ALTER TABLE PARTICIPATION
MODIFY Participation_Date DATE DEFAULT (CURRENT_DATE);
```



Figure 5.3: Table modification

6.Querying in MySQL

At this stage, useful reports using querying are applied for Manufacturing Community club to help them with decision making processes. This will include helping better organize events, monitoring member, attending events, and event data, and analyzing information to improve the management of future activities within the club.

6.1 SQL Reports

Report 1: Events with Highest Attendance

The project manager within operation team needs this report to identify the events that attracted the largest number of attendees to decide which events should be repeated in the future. It also will help Financial Manager to decide the budget for similar events.

```
SELECT EVENT.Event_ID, EVENT.Event_Name,
COUNT(EVENT_ATTENDEE.Attendee_SSN) AS Total_Attendees
FROM EVENT
JOIN EVENT_ATTENDEE
ON EVENT.Event_ID = EVENT_ATTENDEE.Event_ID
GROUP BY EVENT.Event_ID, EVENT.Event_Name
ORDER BY Total_Attendees DESC;
```

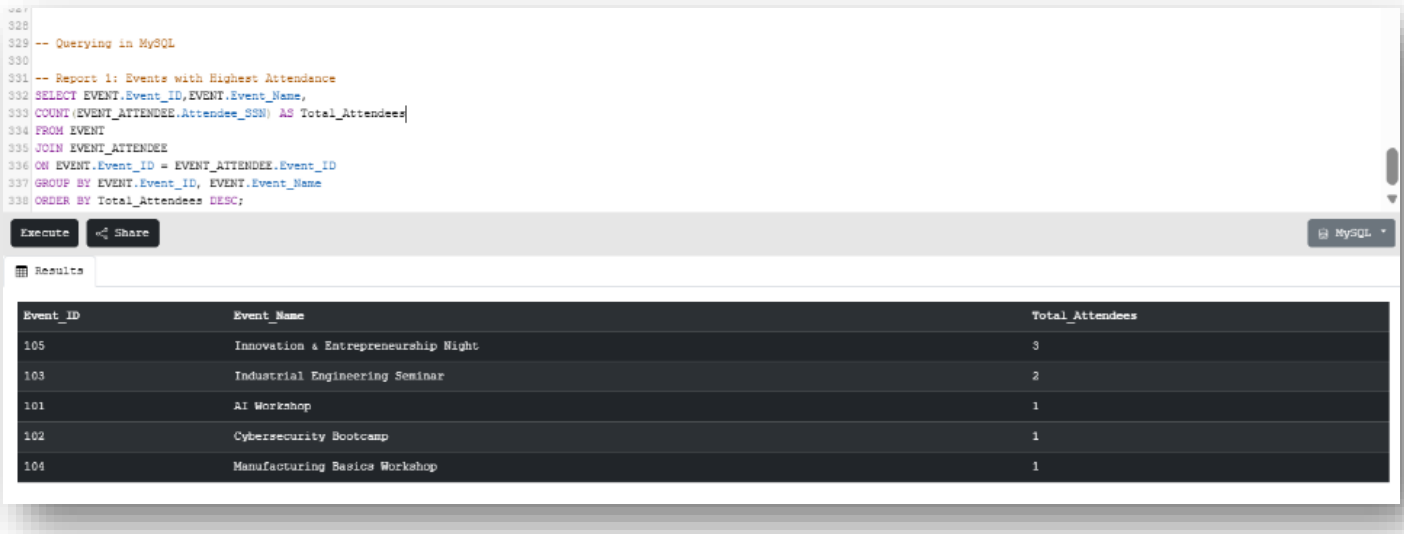


Figure 6.1.1 Report1

Report 2: Total Members per Department Report

Create a report that shows the population of each department by counting the students assigned to it. helping administration identify which departments are the largest and manage workload distribution or space allocation effectively.

```
SELECT Dep_Name, COUNT(Student_ID) AS Total_Members
FROM DEPARTMENT
INNER JOIN MEMBER ON DEPARTMENT.Dep_ID = MEMBER.Dep_ID
GROUP BY Dep_Name
ORDER BY Total_Members DESC;
```

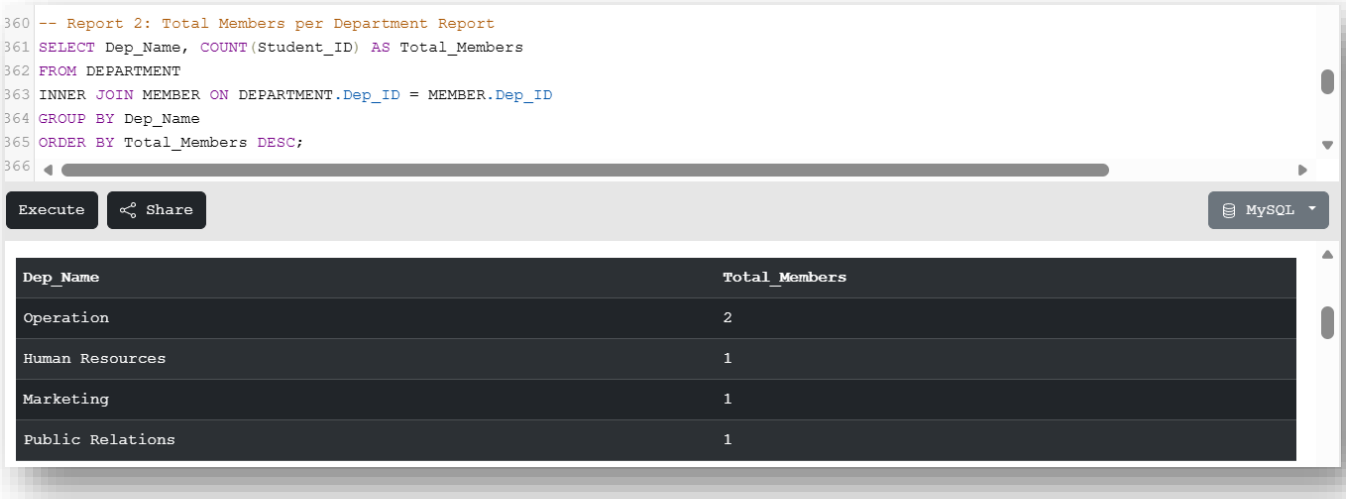


Figure 6.1.2 Report2

Report 3: Events held in June

The Operations Team needs this report to analyze events held at June to evaluate monthly performance.

```
SELECT Event_Name, Event_Date FROM EVENT
WHERE Event_Date BETWEEN '2026-06-01' AND '2026-06-30'
ORDER BY Event_Date;
```

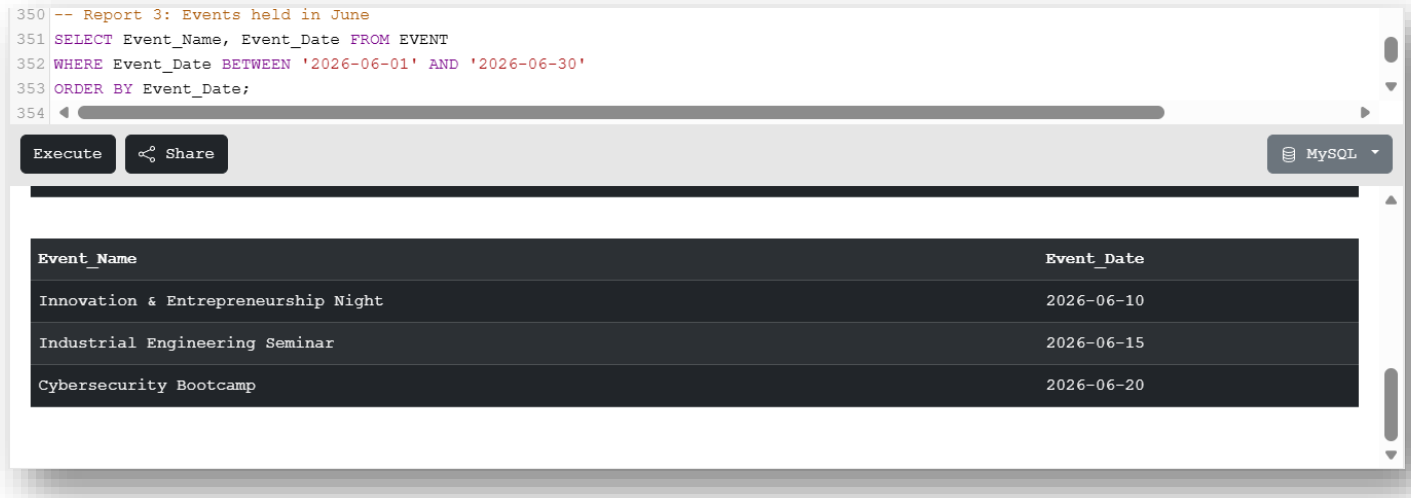


Figure 6.1.3 Report 3

Report 4: Instructor Contact List

The Public Relations Team needs quick access to instructor contact details for coordination of upcoming events and it should be ordered alphabetically by instructors name to make the search and selection easier.

```
SELECT Instructor_Full_Name,Instructor_Email, In_Phone_Number
FROM INSTRUCTOR
ORDER BY Instructor_Full_Name;
```

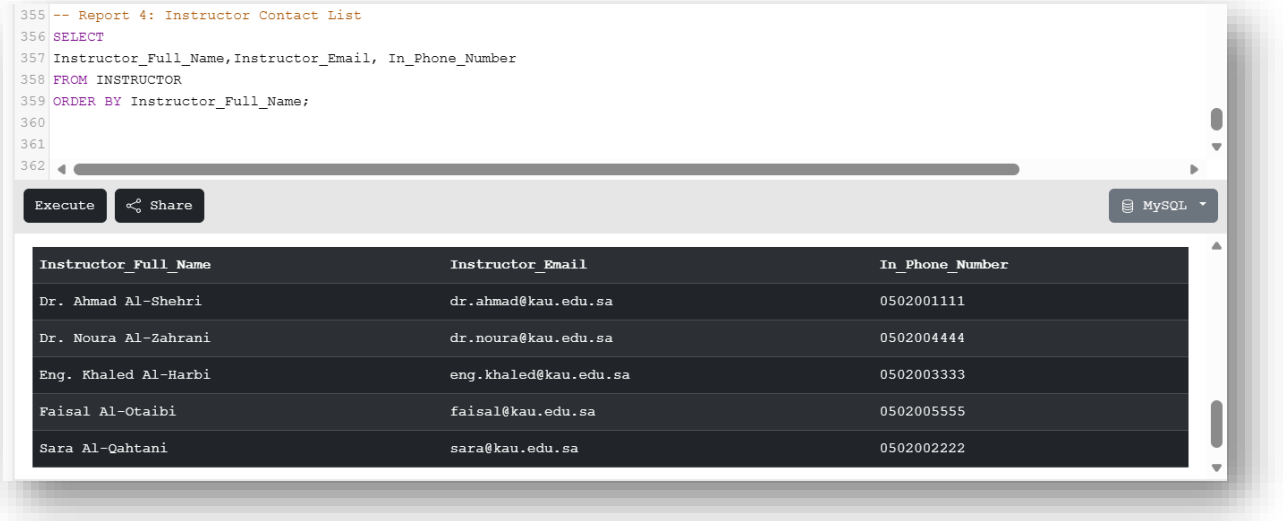


Figure 6.1.4 Report 4

Report 5: Awards for Best Department Based on Achievements

The HR Department needs this report to evaluate department performance based on achievements. This helps in identifying the most active and successful department for recognition, rewards.

```
SELECT Dep_Name, COUNT(A_No) AS Total_Achievements
FROM DEPARTMENT D
JOIN DEPARTMENT_ACHIEVEMENT DA
ON D.Dep_ID = DA.Dep_ID
GROUP BY Dep_Name
ORDER BY Total_Achievements DESC;
```

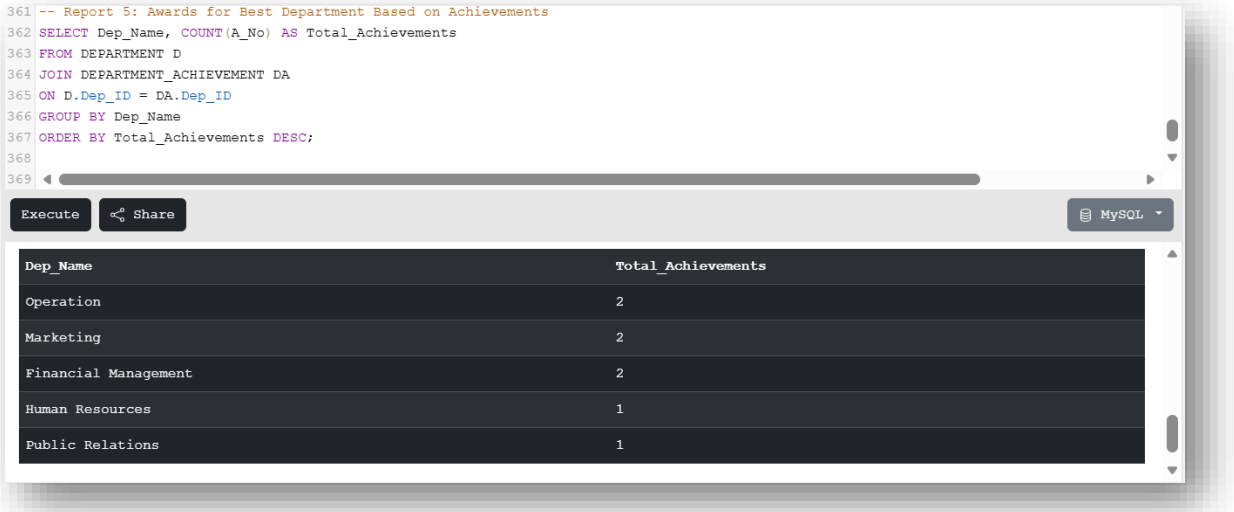


Figure 6.1.5 Report 5

Report 6: Feedback Review and handling Complaints

This report helps the HR team review all complaints received daily to identify urgent issues and take immediate actions to improve event quality.

```
SELECT
Feedback_ID,Feedback_Name,Feedback_Type,Description
FROM FEEDBACK
WHERE Feedback_Type = 'Complaint'
ORDER BY Feedback_ID DESC;
```

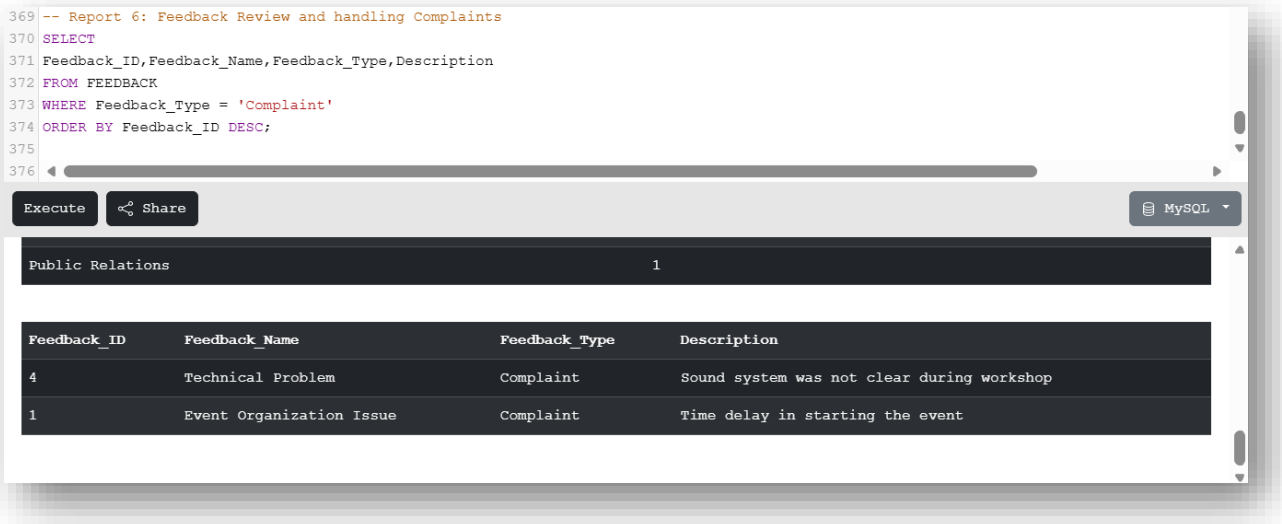


Figure 6.1.6 Report 6

Report 7: Member Participation Performance

The Club manager needs this report to evaluate member engagement in events. This helps in recognizing active members and assigning responsibilities to highly involved ones.

```
SELECT CONCAT(MEMBER.M_FName, ' ', MEMBER.M_LName) AS Member_Name,
COUNT(P.Event_ID) AS Total_Participations,
(COUNT(P.Event_ID) * 100.0 / (SELECT COUNT(*) FROM EVENT)) AS
Participation_Percentage
FROM MEMBER
JOIN PARTICIPATION P
ON MEMBER.Student_ID = P.Student_ID
GROUP BY MEMBER.Student_ID, MEMBER.M_FName,MEMBER.M_LName,
MEMBER.Department
ORDER BY Participation_Percentage DESC;
```

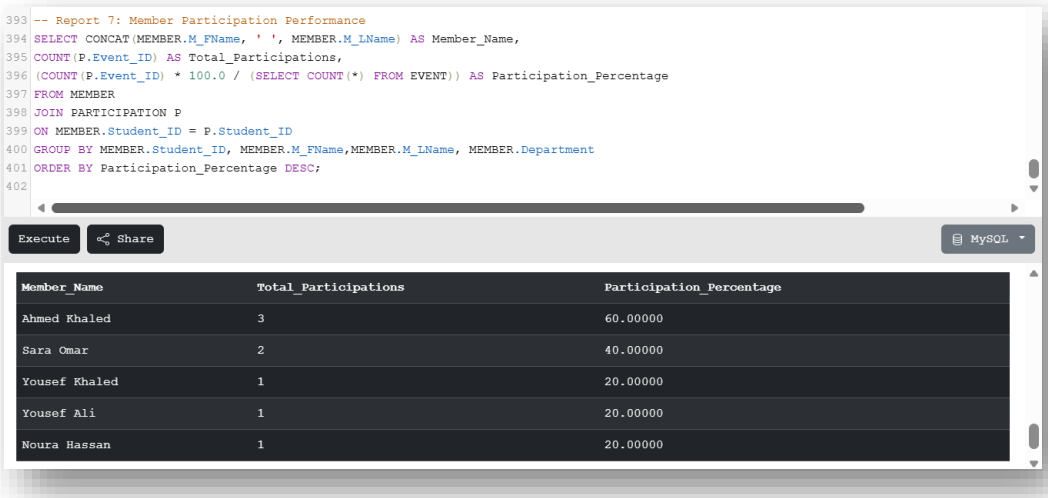


Figure 6.1.7 Report 7

6.2 SQL Data Removal Statement

Delete a sponsor and all their event associations

```
DELETE FROM SPONSOR
WHERE Sponsor_ID = 4001;
SELECT * FROM SPONSOR;
```

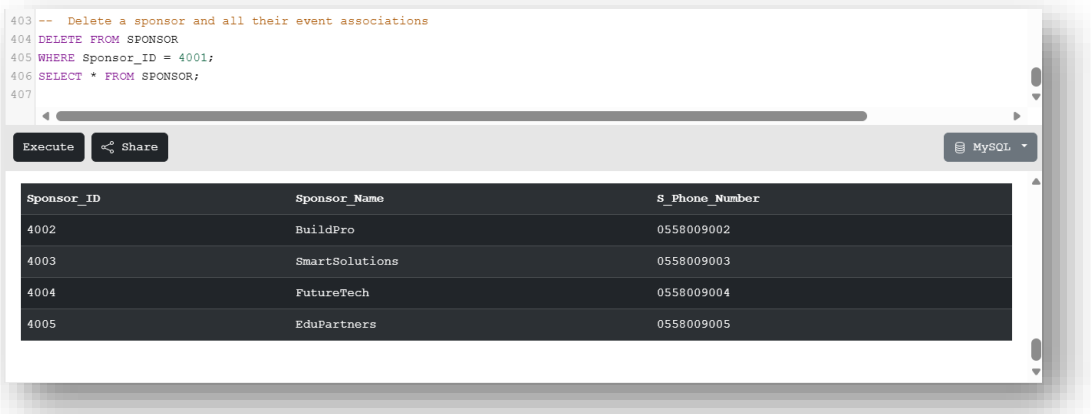


Figure 6.2.1 Remove data from table

Figures 6.2.2 and 6.2.3 below indicate the tables before deleting the data from them

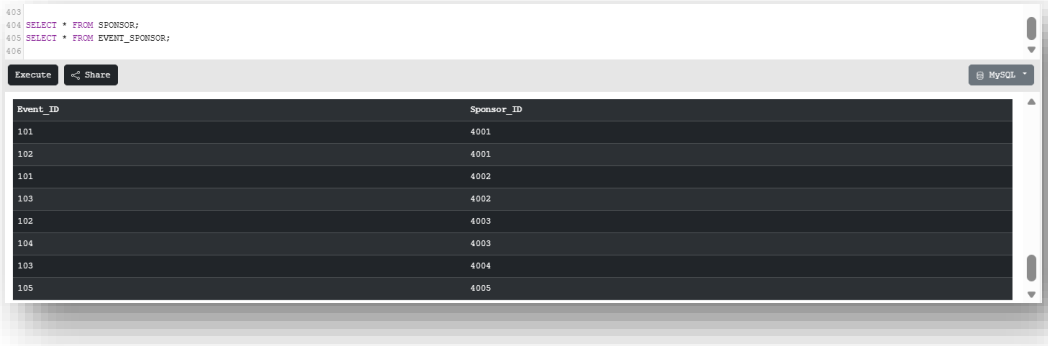


Figure 6.2.2 Event-Sponsor Before remove data

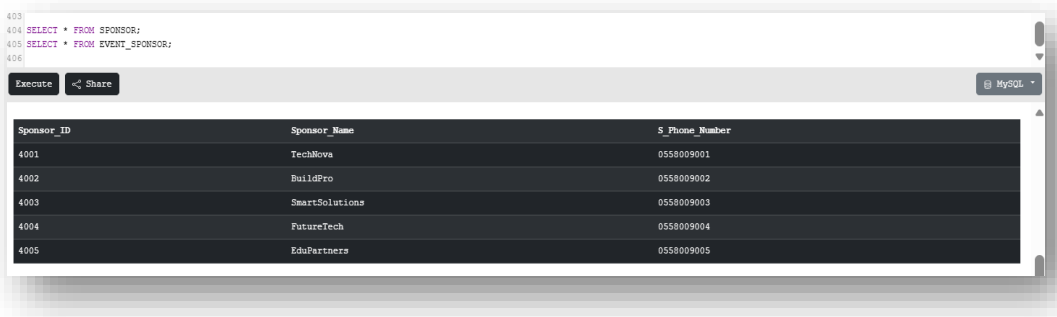


Figure 6.2.3 Sponsor Before remove data

Figures 6.2.4 and 6.2.5 below indicate the tables after deleting the data from them all records that include the ID 4001 was deleted with their associative one in the Event-Sponsor table

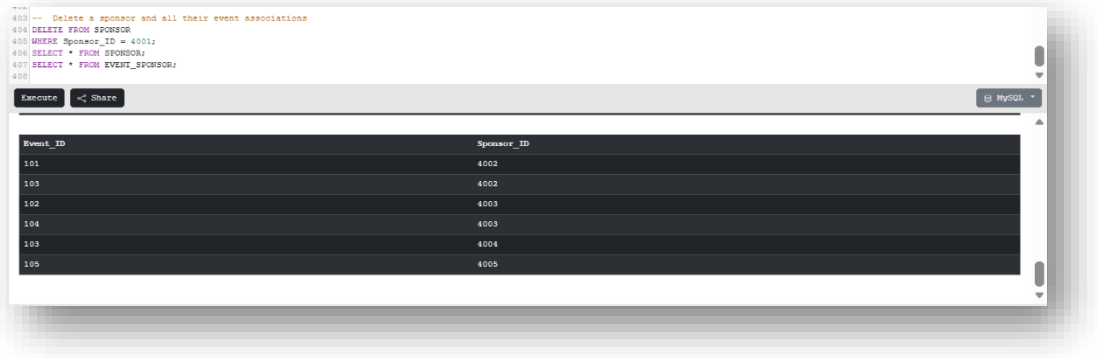


Figure 6.2.4 Event-Sponsor After remove data

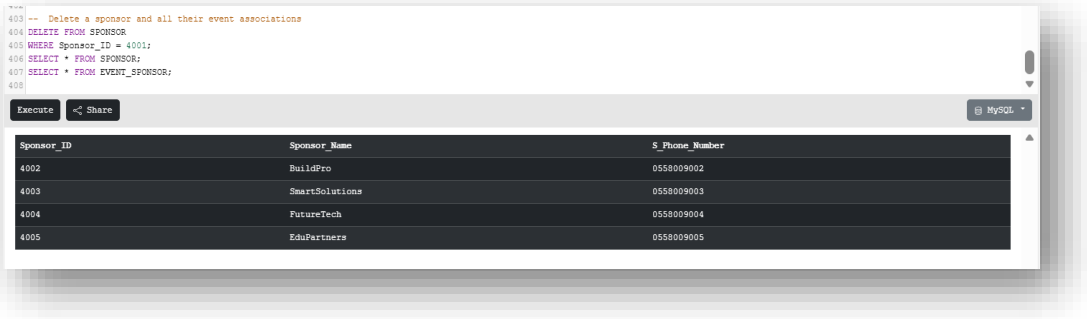
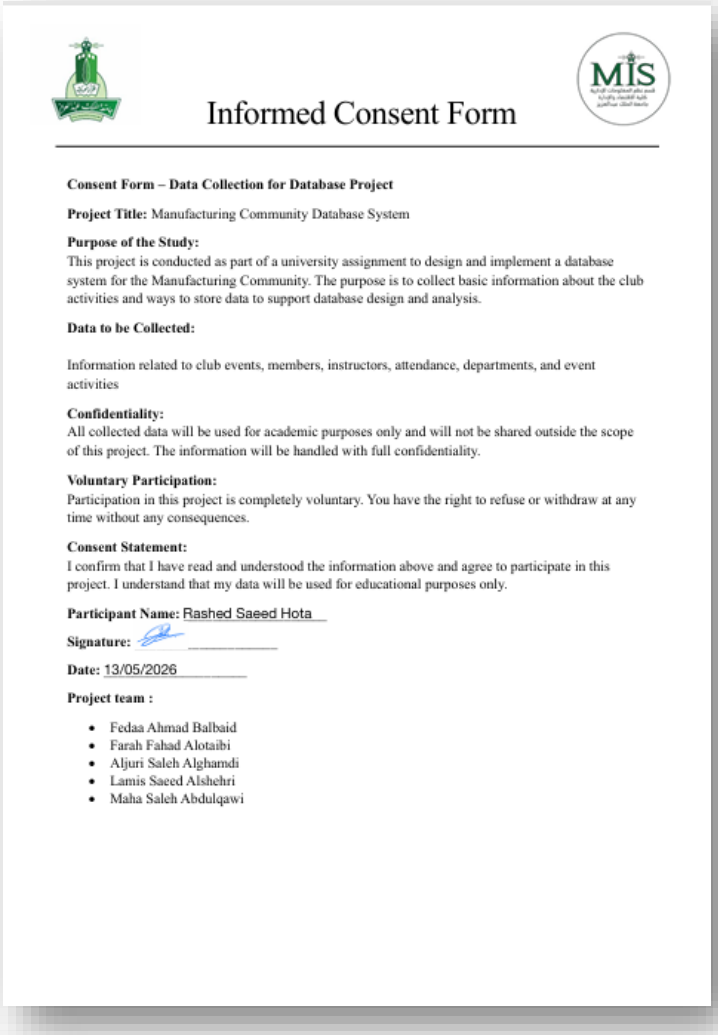


Figure 6.2.5 Sponsor After remove data

Appendix



Informed Consent Form

Consent Form – Data Collection for Database Project

Project Title: Manufacturing Community Database System

Purpose of the Study:
This project is conducted as part of a university assignment to design and implement a database system for the Manufacturing Community. The purpose is to collect basic information about the club activities and ways to store data to support database design and analysis.

Data to be Collected:

Information related to club events, members, instructors, attendance, departments, and event activities

Confidentiality:
All collected data will be used for academic purposes only and will not be shared outside the scope of this project. The information will be handled with full confidentiality.

Voluntary Participation:
Participation in this project is completely voluntary. You have the right to refuse or withdraw at any time without any consequences.

Consent Statement:
I confirm that I have read and understood the information above and agree to participate in this project. I understand that my data will be used for educational purposes only.

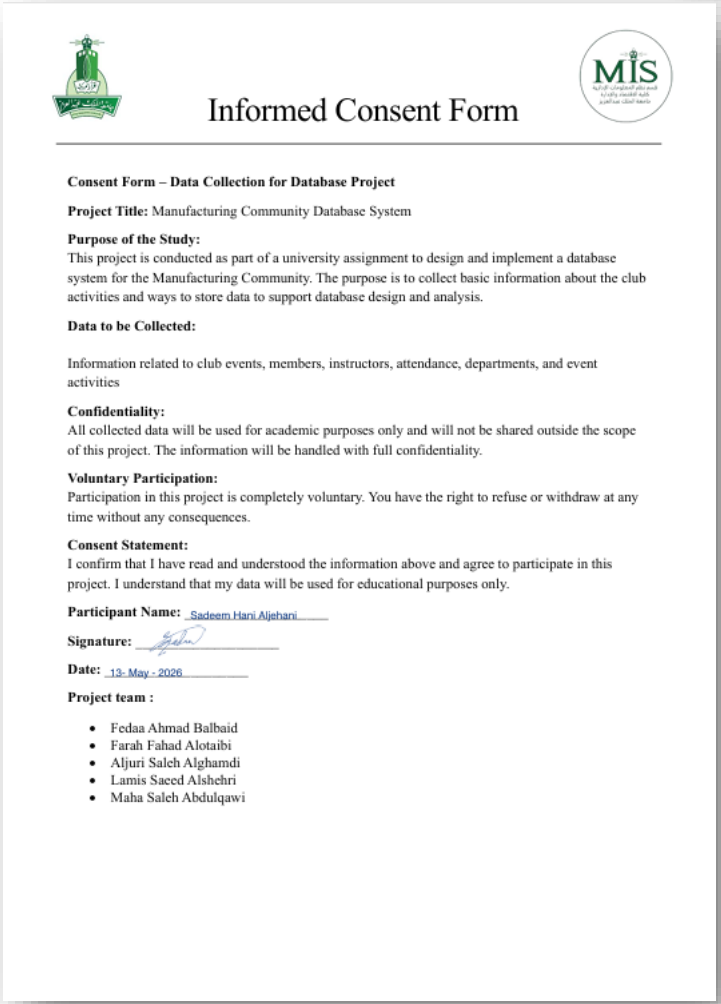
Participant Name: Rashed Saeed Hota

Signature: _____

Date: 13/05/2026

Project team :

- Fedaa Ahmad Balbaid
- Farah Fahad Alotaibi
- Aljuri Saleh Alghamdi
- Lamis Saeed Alshehri
- Maha Saleh Abdulqawi



Informed Consent Form

Consent Form – Data Collection for Database Project

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Consent Statement:
I confirm that I have read and understood the information above and agree to participate in this project. I understand that my data will be used for educational purposes only.

Participant Name: Sadeem Hani Aljehani

Signature: _____

Date: 13- May - 2026

Project team :

- Fedaa Ahmad Balbaid
- Farah Fahad Alotaibi
- Aljuri Saleh Alghamdi
- Lamis Saeed Alshehri
- Maha Saleh Abdulqawi